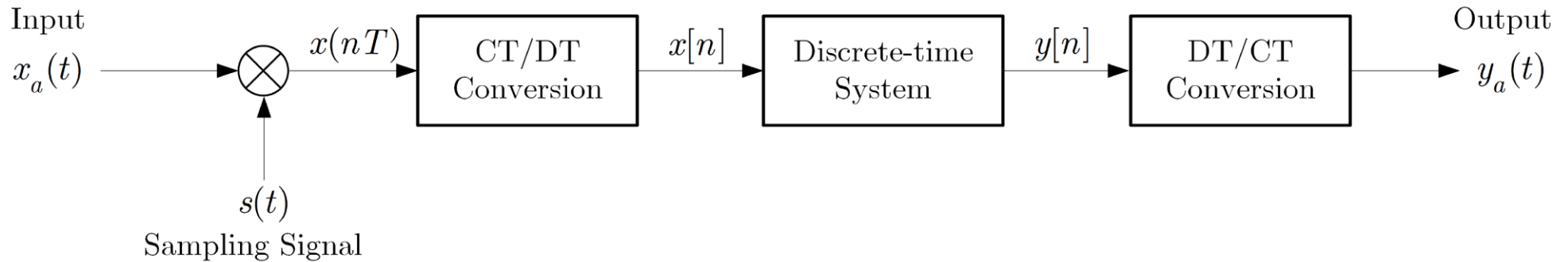


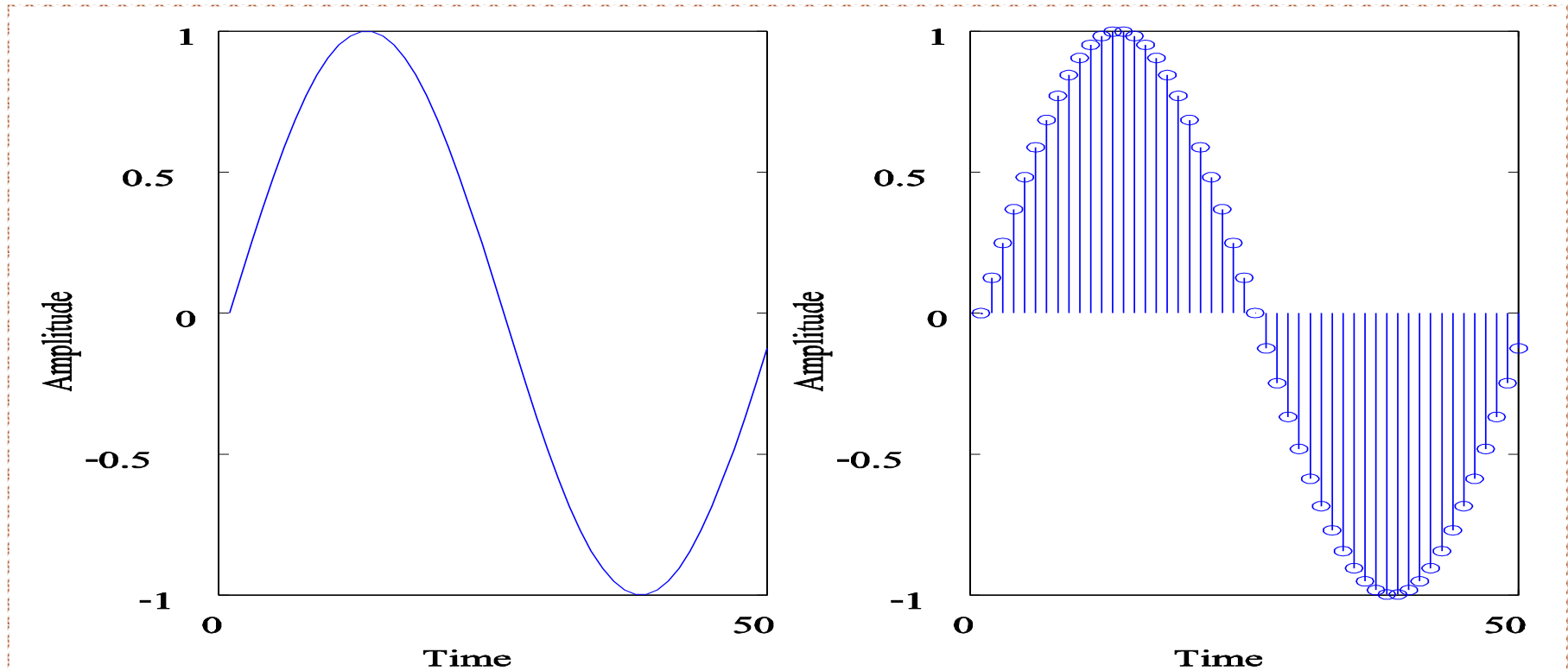
Objectives

- The students understand the concepts of:
 - Discrete-time signals and systems
 - Linear Time-Invariant Systems
 - Convolution

Digital Signal Processing Building Blocks

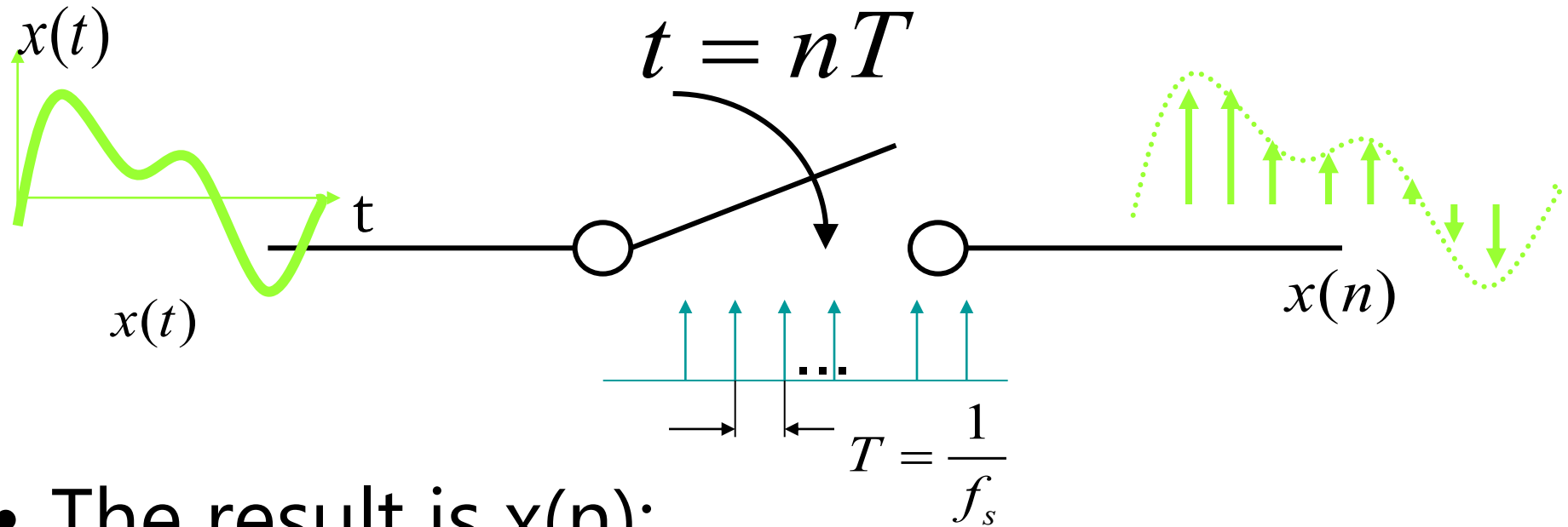


Continuous vs. Discrete-time Signals



Sampling

- We get the digital signal by sampling

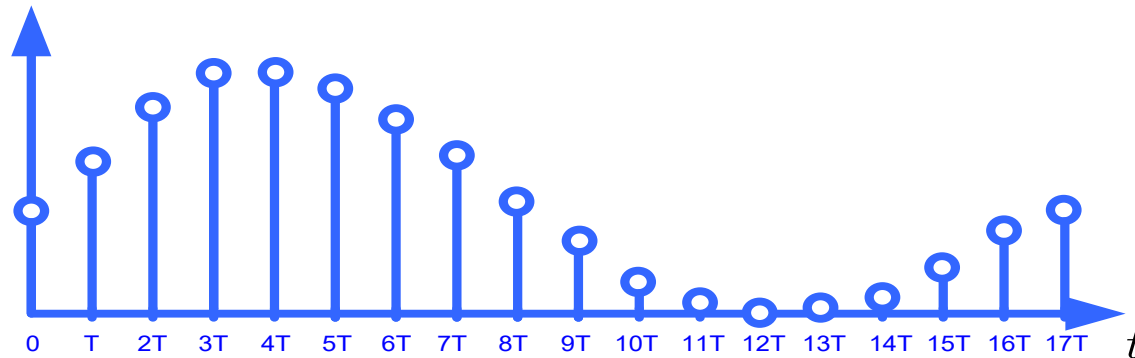


- The result is $x(n)$:

$$x(n) = x(t) \Big|_{t=nT}$$

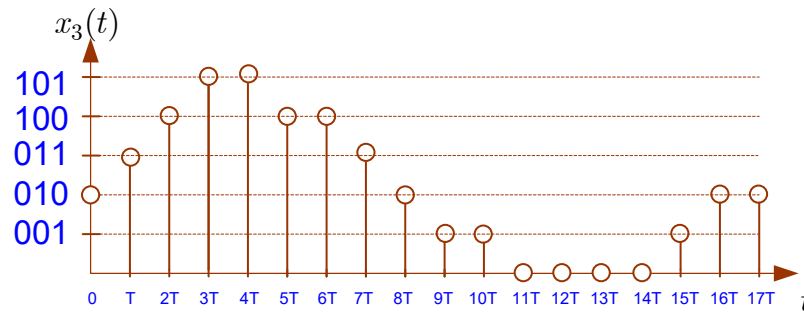
Discrete-Time Continuous Amplitude

- We are only interested in **Discrete-Time, Continuous Amplitude** signals.

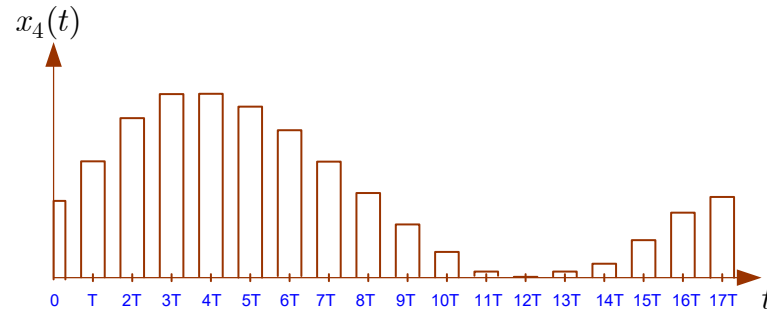


Discrete time ,
Continuous Amplitude

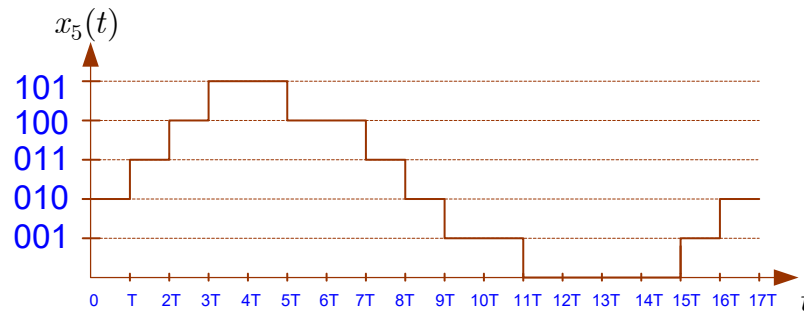
Other Types of Signals



Discrete time,
Discrete Amplitude



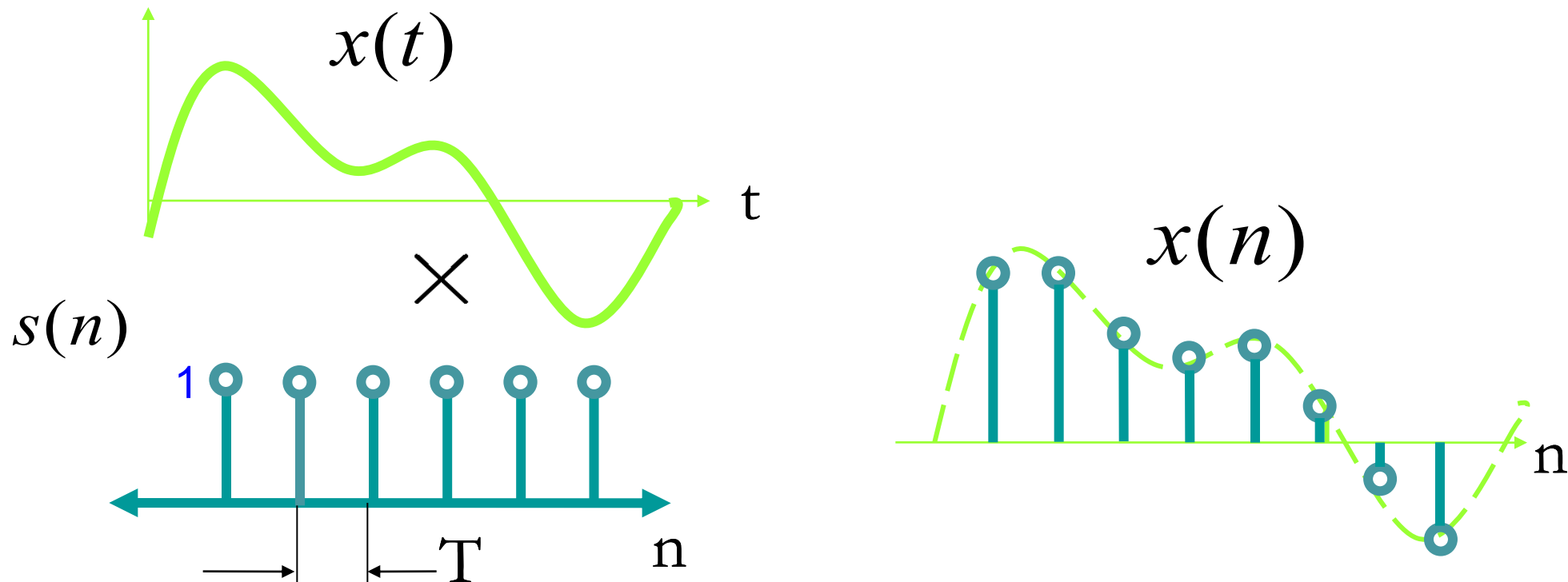
Continuous time,
Continuous Amplitude,
Uniform time steps



Continuous time,
Discrete Amplitude
Uniform time steps

Discrete-time signals obtained by Sampling

Discrete-time " $x(n)$ " = Sampling " $s(n)$ " x Analogue " $x(t)$ "



- $S(n)$ is composed of **delayed Impulses**

An Impulse is Delta Function

$$\delta(n) = \delta(t) \Big|_{t=nT}$$



- The definition of impulse is

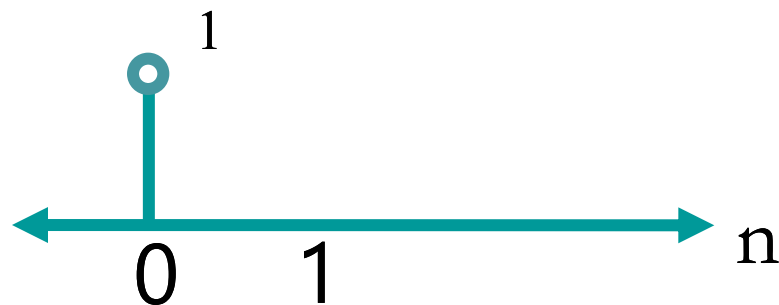
$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Delayed Delta Functions

- The unit delayed delta function

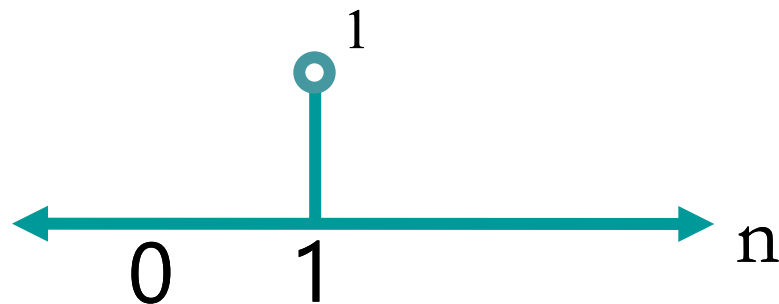
Non-delayed impulse

$$\delta(n)$$



The unit-delayed impulse

$$\delta(n-1)$$



Summing of Delayed Delta Functions

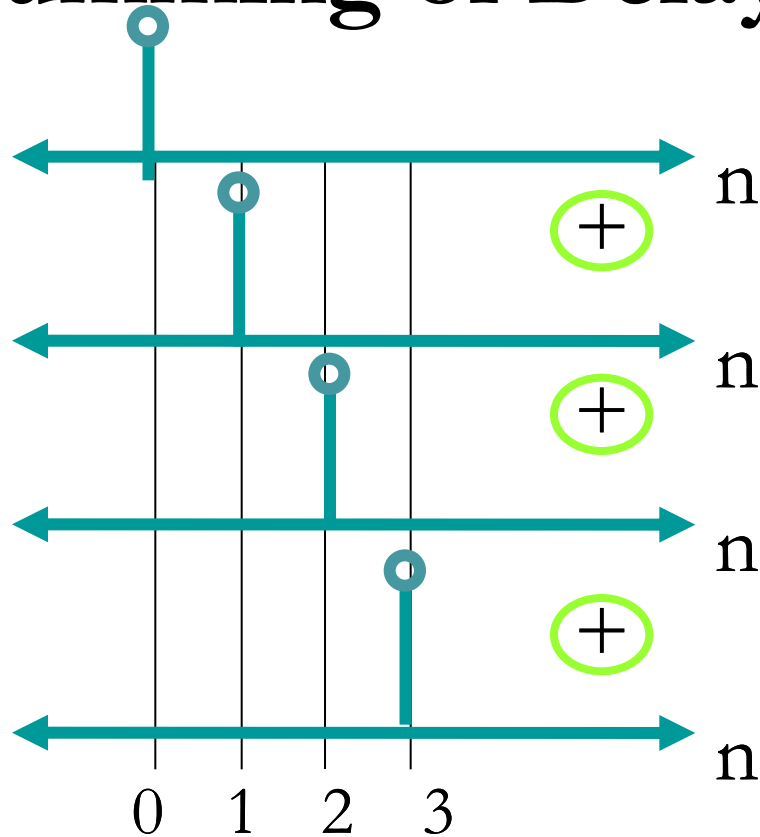
- Sampling signal can be derived as a sum of delayed impulses. For example, for $N=3$

$$s(n) = \delta(n) + \delta(n-1) + \delta(n-2) + \delta(n-3)$$

- Or

$$s(n) = \sum_{k=0}^3 \delta(n-k)$$

Summing of Delayed Delta Functions



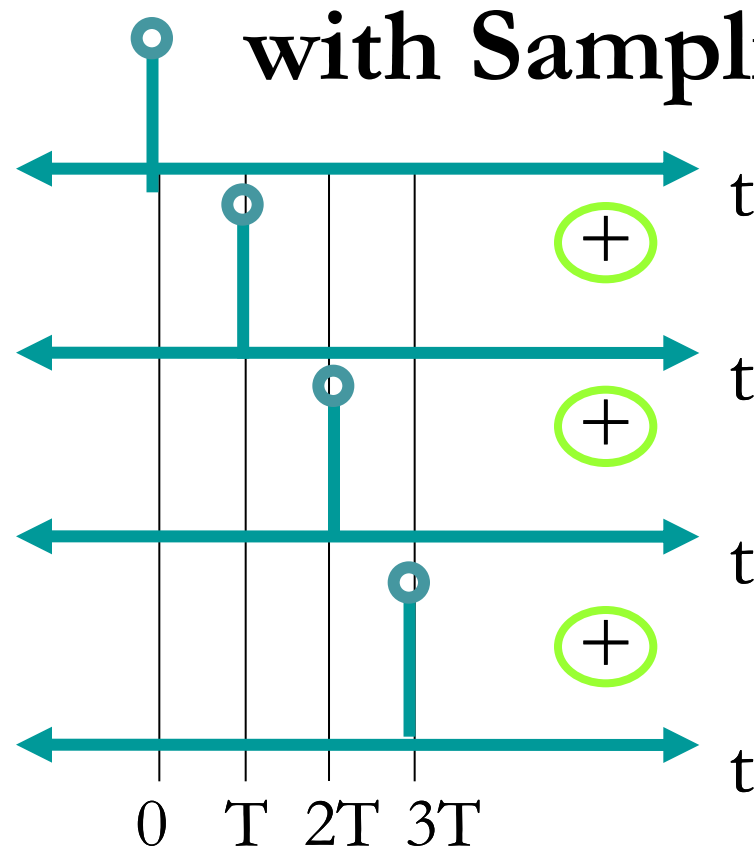
$$\begin{aligned} & \delta(n) \\ & + \\ & \delta(n-1) \\ & + \\ & \delta(n-2) \\ & + \\ & \delta(n-3) \end{aligned}$$

=

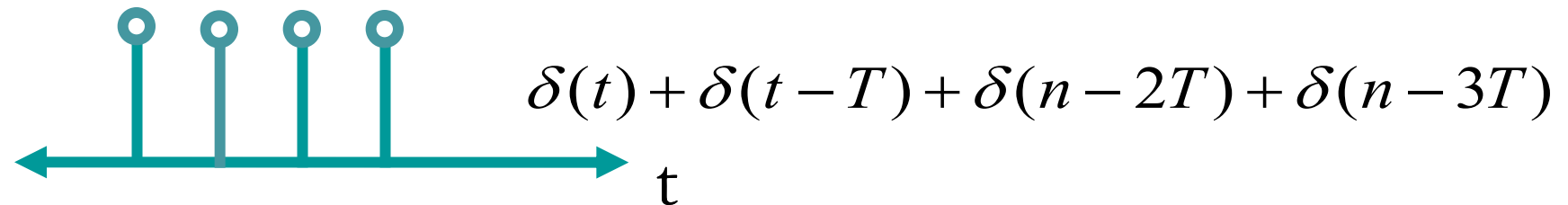


$$\delta(n) + \delta(n-1) + \delta(n-2) + \delta(n-3)$$

Summing of Delayed Delta Functions

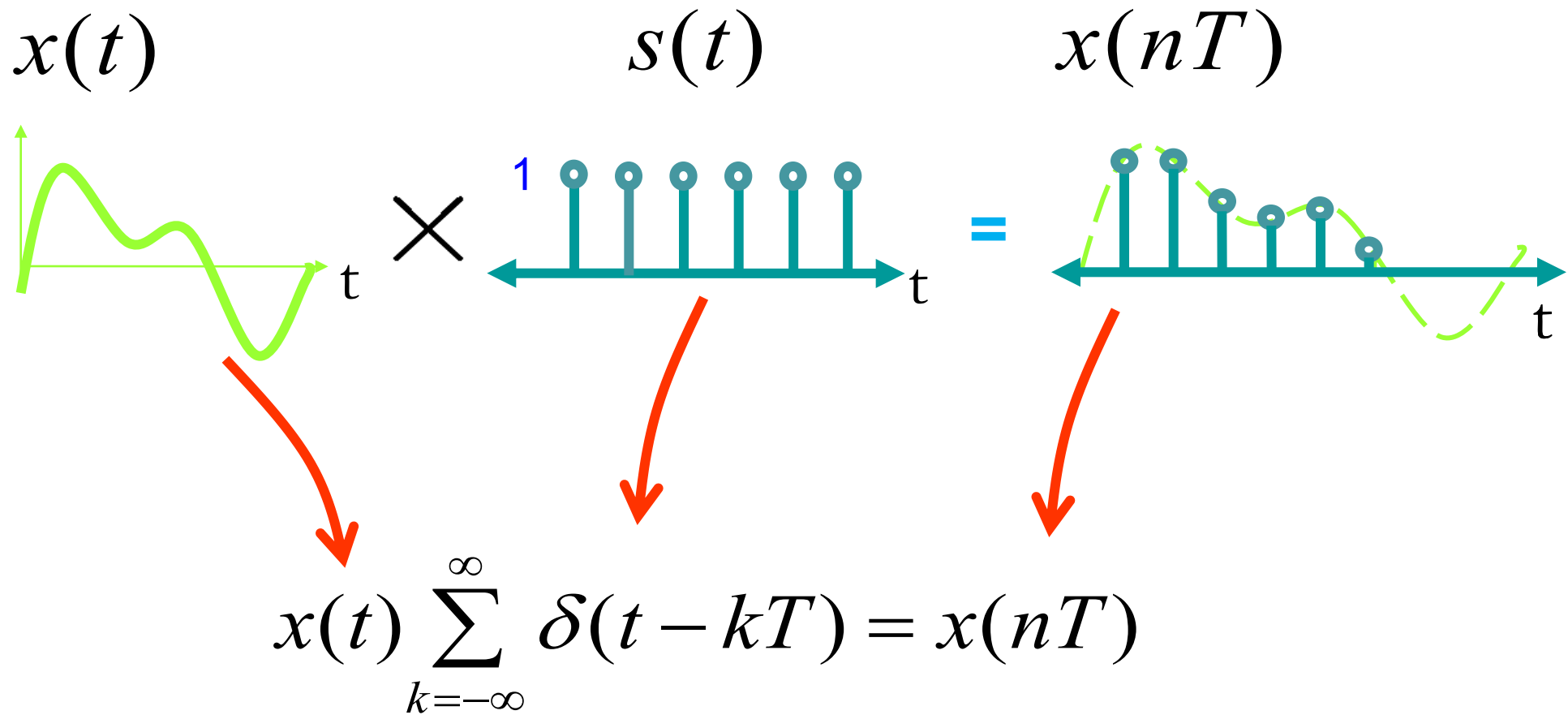


$$\begin{aligned}
 & \delta(t) \\
 & + \\
 & \delta(t - T) \\
 & + \\
 & \delta(t - 2T) \\
 & + \\
 & \delta(t - 3T)
 \end{aligned}
 =$$



Continuous-time to Discrete-time

$x(nT)$



Converting Continuous-time to Discrete-time

- From

$$x(t) \sum_{k=-\infty}^{\infty} \delta(t - kT) = x(nT)$$

- Let $x(t)=x(kT)$:

$$x(kT) \sum_{k=-\infty}^{\infty} \delta(t - kT) = x(nT)$$

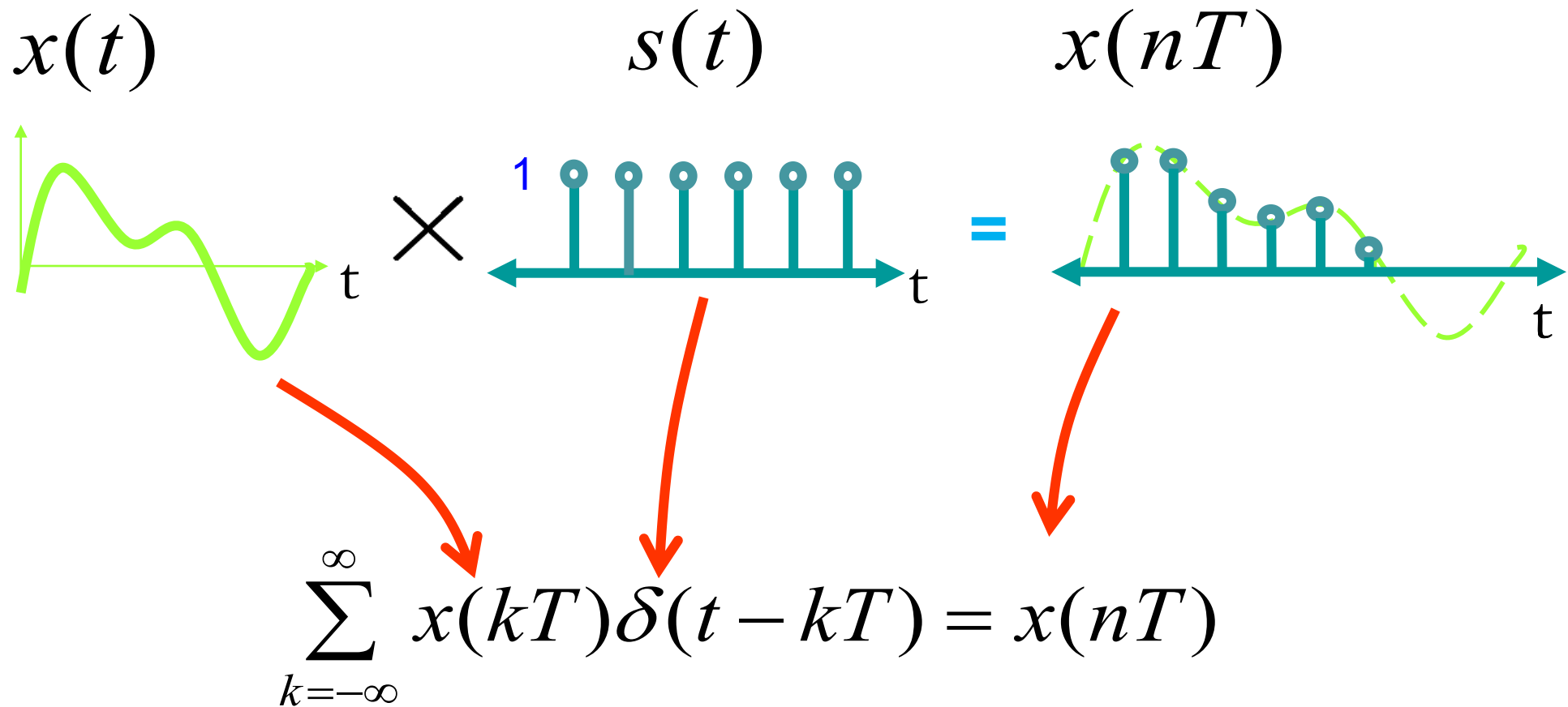
- Move $x(kT)$

Inside the sum:

$$\sum_{k=-\infty}^{\infty} x(kT) \delta(t - kT) = x(nT)$$

Continuous-time to Discrete-time

$x(nT)$

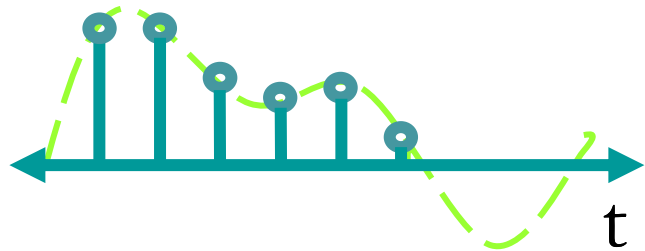


Convert to a sequence

- We can drop the sampling interval T by to make a sequence $x(n)$

$$x(nT) \Rightarrow x(n)$$

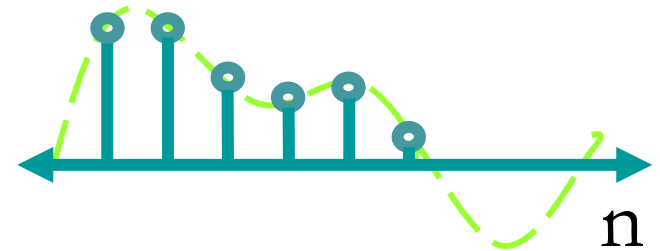
$x(nT)$



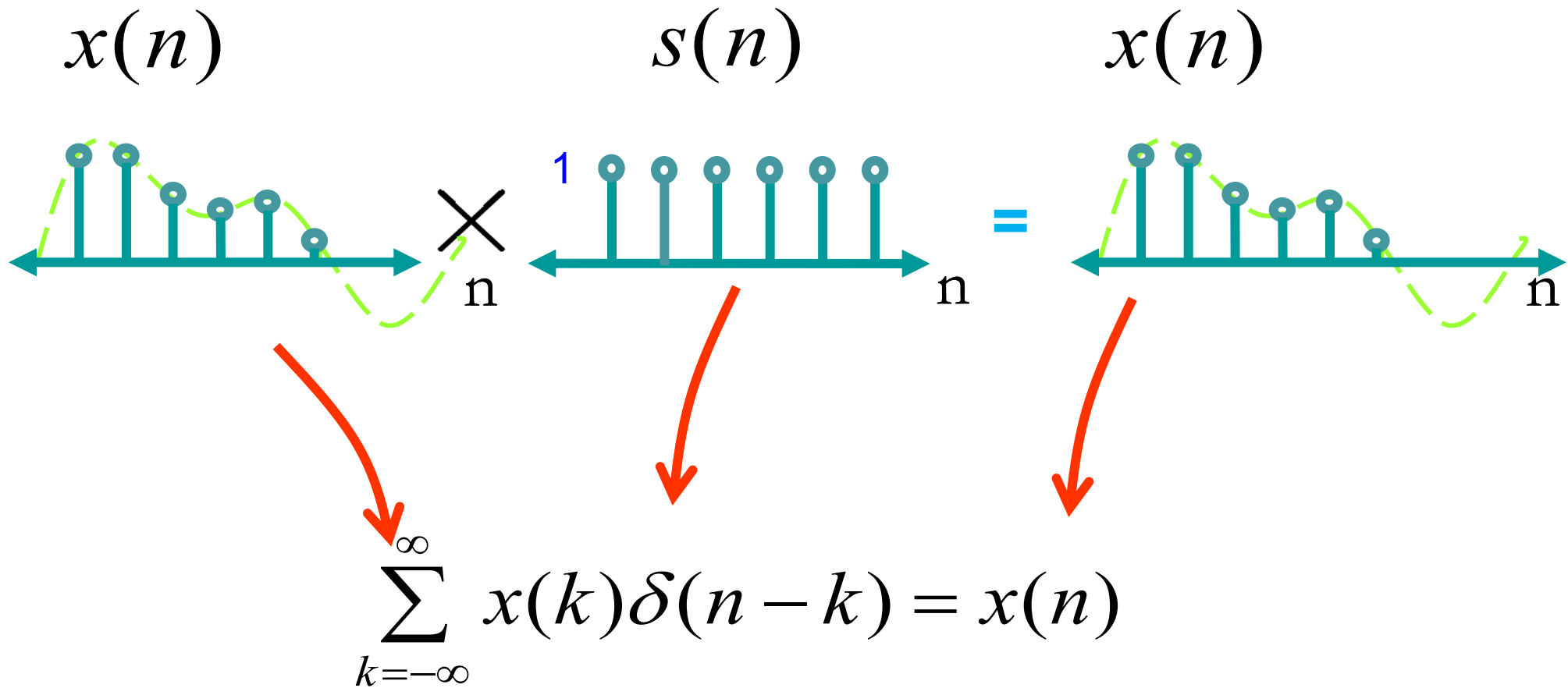
T is normalised



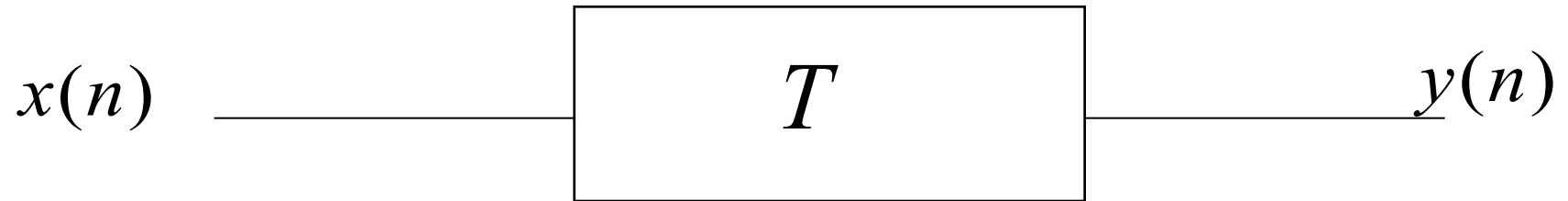
$x(n)$



Discrete-time Signal $x(n]$ from $x(n]$

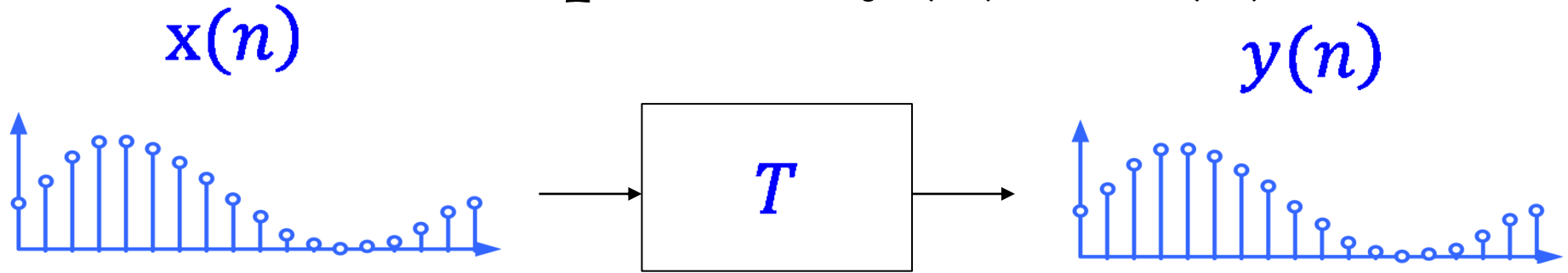


Discrete-Time Systems



- $x(n)$ is the input of the system T
- $y(n)$ is the output of the system T
- T is the system and is characterised by its **impulse response**.

Example 2.0: $y(n) = x(n)$



$$y(0) = x(0)$$

$$y(1) = x(1)$$

$$y(2) = x(2)$$

$$\vdots$$

$$y(10) = x(10)$$

Example 2.1: System 1

- Example 2.2.1
$$x(n) = \begin{cases} |n|, & -3 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

- Determine $y(n)$ when

A) $y(n) = x(n)$

B) $y(n) = x(n-1)$

C) $y(n) = x(n+1)$

D) $y(n) = \frac{1}{3}[x(n+1) + x(n) + x(n-1)]$

E) $y(n) = \max\{x(n+1), x(n), x(n-1)\}$

F) $y(n) = \sum_{k=-\infty}^n x(k)$

Example 2.1: System

$$A) \quad y(n) = x(n)$$

$$y(n) = x(n) = \left\{ \quad , 3, 2, 1, \underset{\uparrow}{0}, 1, 2, 3, \quad \right\}$$

- Note: the symbol $\uparrow$ refers to the position when $n=0$.

Example 2.1: System

$$B) \quad y(n) = x(n-1)$$

$$y(n) = \left\{ \quad , 3, 2, \underset{\uparrow}{1}, 0, 1, 2, 3, \quad \right\}$$

Example 2.1: System

$$C) \quad y(n) = x(n+1)$$

$$y(n) = \left\{ \begin{array}{l} \dots \\ \dots \end{array} \right.$$

Example 2.1: System

$$D) \quad y(n) = \frac{1}{3} [x(n+1) + x(n) + x(n-1)]$$

$$n = 0, \quad y(0) = \frac{1}{3} [x(-1) + x(0) + x(1)] = \frac{1}{3} [1 + 0 + 1] = \frac{2}{3}$$

$$y(n) = \{ \dots, 4, 3, 2, 1, \frac{2}{3}, 1, 2, 3, 4, \dots \}$$

↑

Example 2.1: System

$$E) \quad y(n) = \max \{x(n+1), x(n), x(n-1)\}$$

$$y(n) = \{..., 4, 3, 2, \underset{\uparrow}{1}, 2, 3, 4, ...\}$$

Example 2.1 : System 4

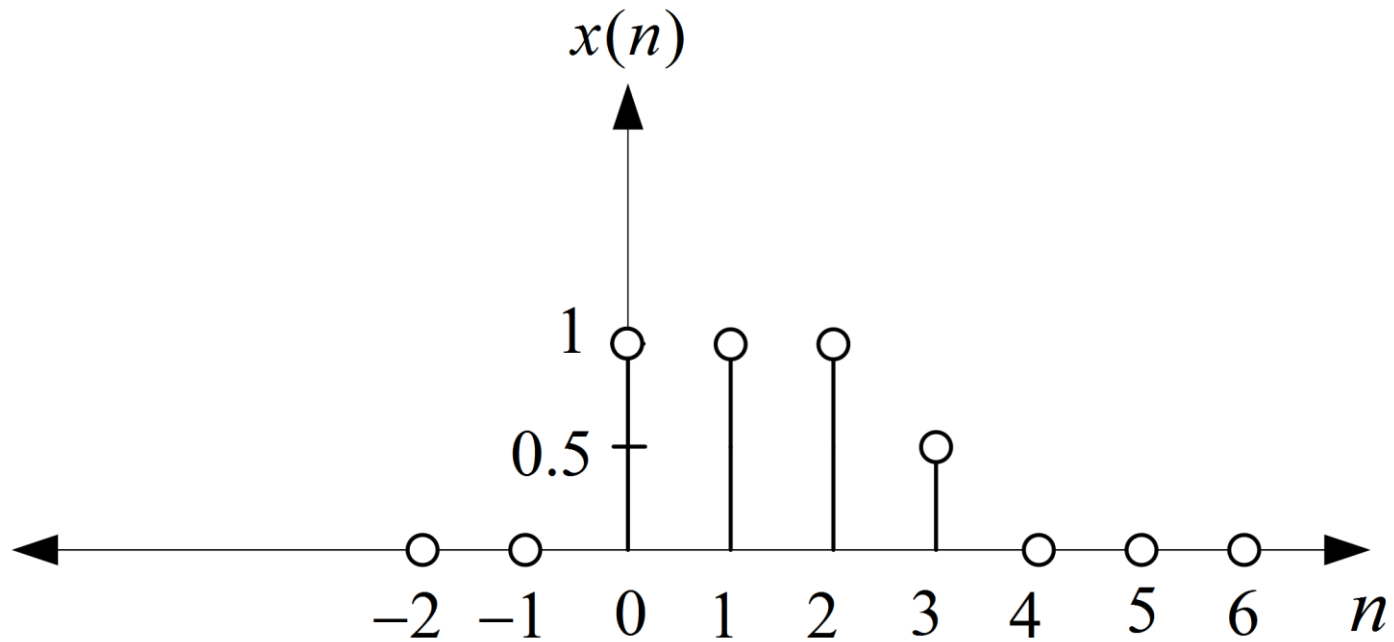
- $F) \quad y(n) = \sum_{k=-\infty}^n x(k)$

- Accumulator

$$y(n) = \left\{ \dots \quad \underset{\uparrow}{5}, 7, 9, 12, 0, \dots \right\}$$

Example 3

- Given $x(n]$



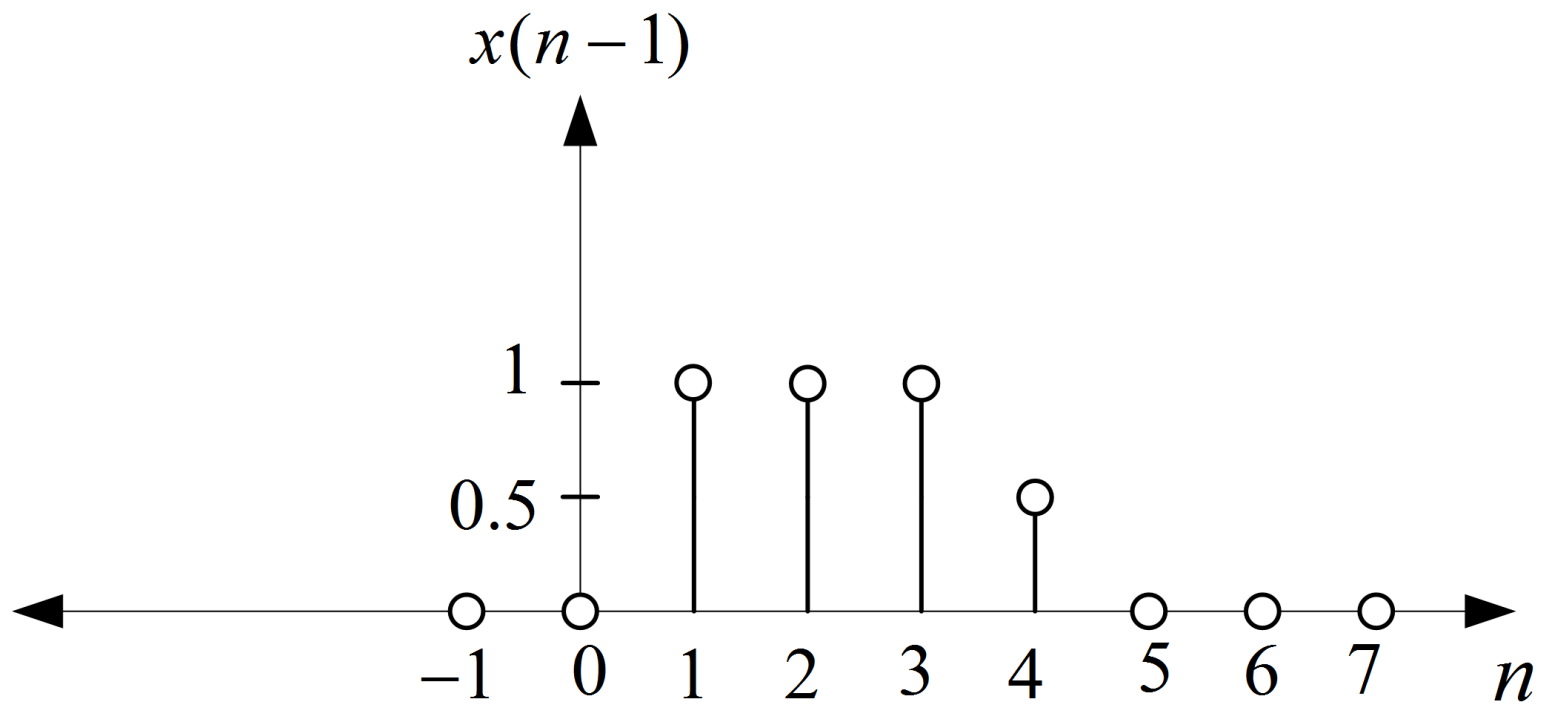
$$\mathbf{x(n)}$$

n	X(n)
-2	0
-1	0
0	1
1	1
2	1
3	0.5
4	0

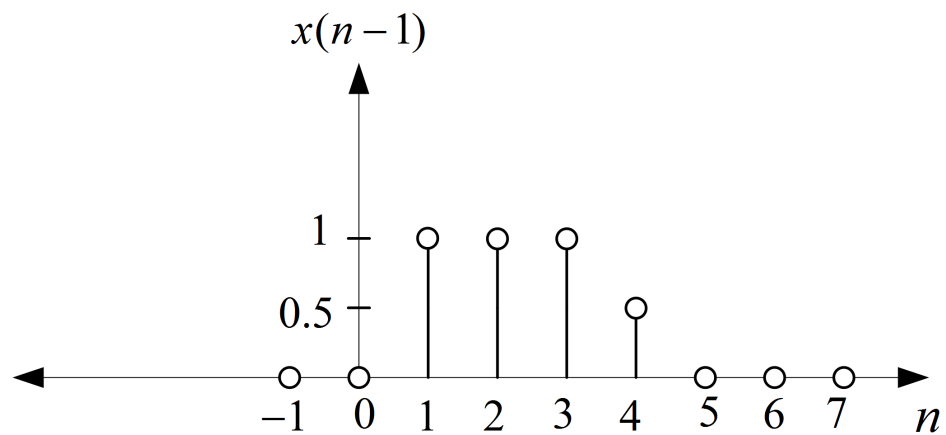
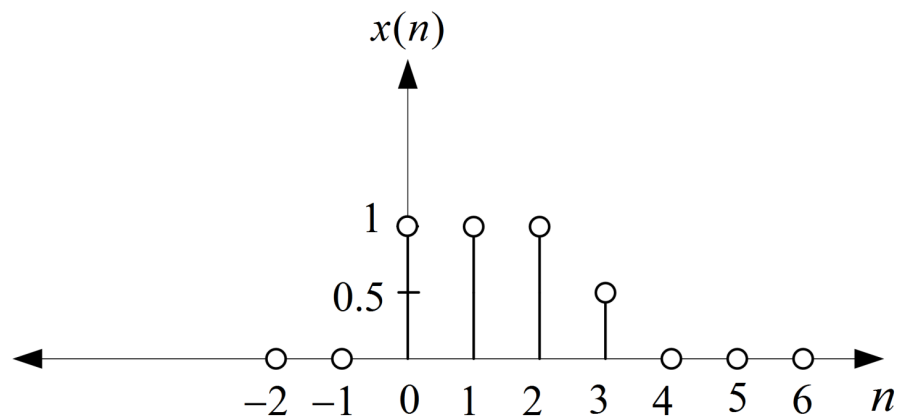
$$x(n-1)$$

n	X(n)	X(n-1)	Result X(n-1)
-2	0	$x(-2-1)=x(-3)$	0
-1	0	$x(-1-1) = x(-2)$	0
0	1	$x(0-1) = x(-1)$	0
1	1	$x(1-1) = x(0)$	1
2	1	$x(2-1) = x(1)$	1
3	0.5	$x(3-1) = x(2)$	1
4	0	$x(4-1)=x(3)$	0.5
5	0	$X(5-1)=x(4)$	0
6	0	$X(6-1)=x(5)$	0

$X(n-1)$



$x(n]$ vs. $x(n-1]$

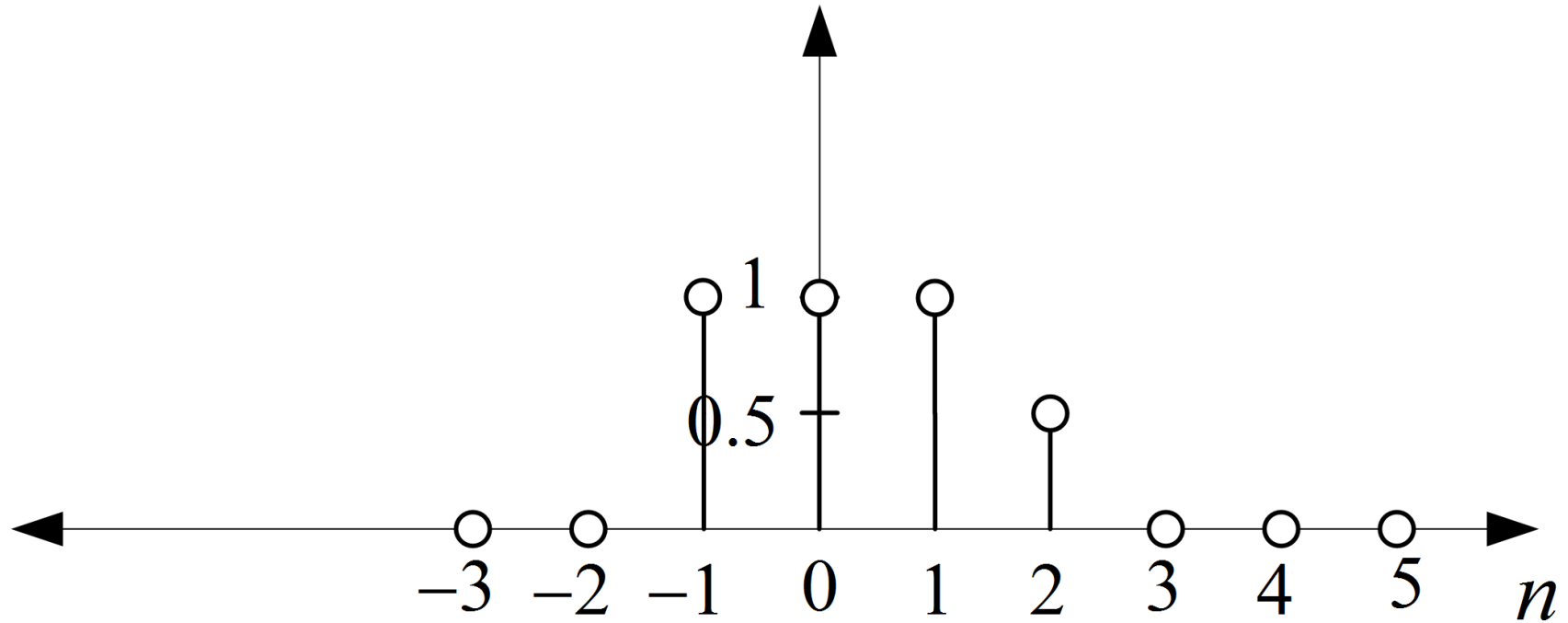


$$\mathbf{x(n+1)}$$

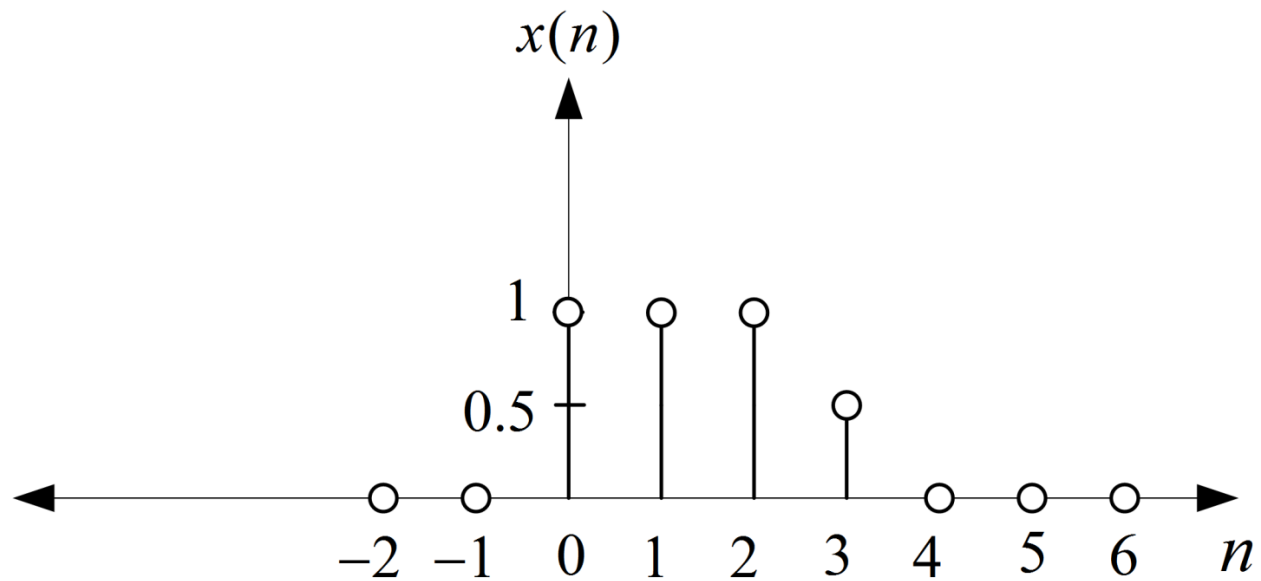
n	X(n)	X(n+1)	Result X(n+1)
-2	0	$x(-2+1)=x(-1)$	0
-1	0	$x(-1+1) =x(0)$	1
0	1	$x(0+1) =x(1)$	1
1	1	$x(1+1) =x(2)$	1
2	1	$x(2+1) =x(3)$	0.5
3	0.5	$x(3+1) =x(4)$	0
4	0	$x(4+1)=x(5)$	0
5	0	$X(5+1)=x(6)$	0
6	0	$X(6+1)=x(7)$	0

$\mathbf{x(n+1)}$

$x(n+1)$



Homework 1



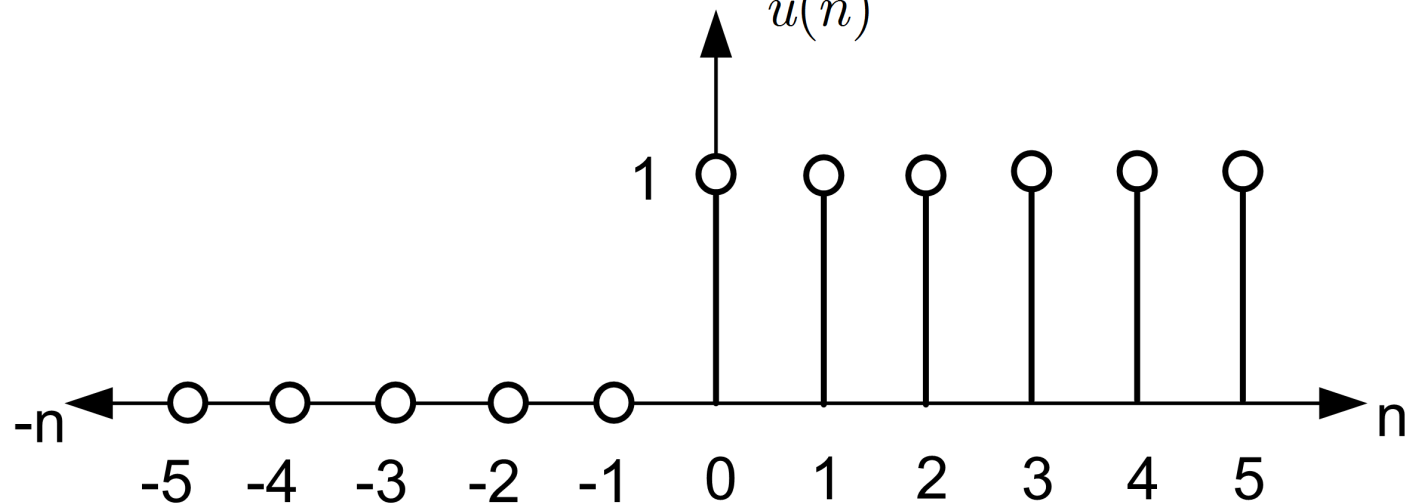
a) $x(-n]$

b) $x(-n+1]$

c) $x(-n-1]$

Unit Step Sequence

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



Relationship between the unit step function and the Delta function

- Define $u(n)$

$$u(n) = \delta(n) + \delta(n-1) + \delta(n-2) + \delta(n-3) + \dots$$

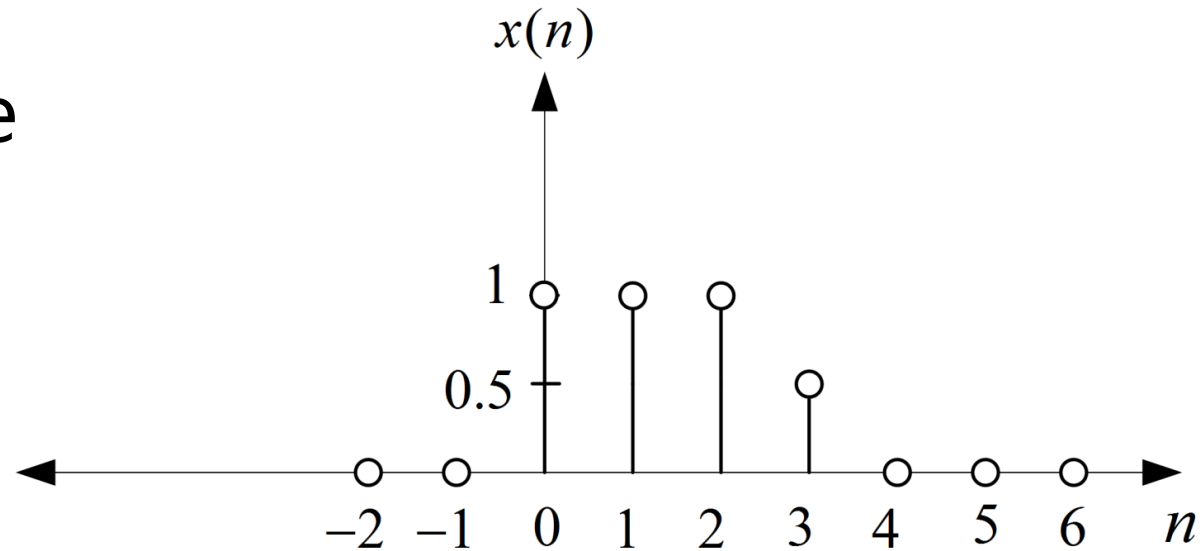
$$= \sum_{k=0}^{\infty} \delta(n-k)$$

Or

$$u(n) = \sum_{k=-\infty}^n \delta(k)$$

Example 4

- Let $x(n]$ be



- Determine

a) $x[n]u[n]$

b) $x[n+1]u[n]$

c) $x[n+1]u[2-n]$

d) $x[-n+1]u[-1-n]$

$$\mathbf{x(n)}$$

n	x(n)
-2	0
-1	0
0	1
1	1
2	1
3	0.5
4	0

$$u(n)$$

n	u(n)
-2	$u(-2)=0$
-1	$u(-1)=0$
0	$u(0)=1$
1	$u(1)=1$
2	$u(2)=1$
3	$u(3)=1$
4	$u(4)=1$

$$x(n)u(n)$$

n	$x(n)u(n)$
-2	0
-1	0
0	1
1	1
2	1
3	0.5
4	0

$$\mathbf{x(n-1)u(n)}$$

n	x(n)	x(n-1)	u(n)	Result x(n-1)u(n)
-2	0	$x(-2-1)=x(-3)$	0	0
-1	0	$x(-1-1) = x(-2)$	0	0
0	1	$x(0-1) = x(-1)$	1	0
1	1	$x(1-1) = x(0)=1$	1	1
2	1	$x(2-1) = x(1)=1$	1	1
3	0.5	$x(3-1) = x(2)=1$	1	1
4	0	$x(4-1)=x(3)=.5$	1	0.5
5	0	$X(5-1)=x(4)$	1	0
6	0	$X(6-1)=x(5)$	1	0

$$\mathbf{x(n+1)u(n)}$$

n	x(n)	x(n+1)	u(n)	Result x(n+1)u(n)
-2	0	$x(-2+1)=x(-1)$	0	0
-1	0	$x(-1+1) = x(0)=1$	0	0
0	1	$x(0+1) = x(1)=1$	1	1
1	1	$x(1+1) = x(2)=1$	1	1
2	1	$x(2+1) = x(3)=.5$	1	0.5
3	0.5	$x(3+1) = x(4)$	1	0
4	0	$x(4+1)=x(5)$	1	0
5	0	$X(5+1)=x(6)$	1	0
6	0	$X(6+1)=x(7)$	1	0

$$u(2-n)$$

n	$u(n)$	$u(2-n)$
-2	$u(-2)=0$	$u(2+2)=1$
-1	$u(-1)=0$	$u(2+1)=1$
0	$u(0)=1$	$u(2-0)=1$
1	$u(1)=1$	$u(2-1)=1$
2	$u(2)=1$	$u(2-2)=1$
3	$u(3)=1$	$u(2-3)=0$
4	$u(4)=1$	$u(2-4)=0$

$$u(2+n)$$

n	u(n)	u(2+n)
-2	$u(-2)=0$	$u(2-2)=1$
-1	$u(-1)=0$	$u(2-1)=1$
0	$u(0)=1$	$u(2-0)=1$
1	$u(1)=1$	$u(2+1)=1$
2	$u(2)=1$	$u(2+2)=1$
3	$u(3)=1$	$u(2+3)=1$
4	$u(4)=1$	$u(2+4)=1$

$$u(-2+n)$$

n	$u(n)$	$u(-2+n)$
-2	$u(-2)=0$	$u(-2-2)=0$
-1	$u(-1)=0$	$u(-2-1)=0$
0	$u(0)=1$	$u(-2-0)=0$
1	$u(1)=1$	$u(-2+1)=0$
2	$u(2)=1$	$u(-2+2)=1$
3	$u(3)=1$	$u(-2+3)=1$
4	$u(4)=1$	$u(-2+4)=1$

$$u(-2-n)$$

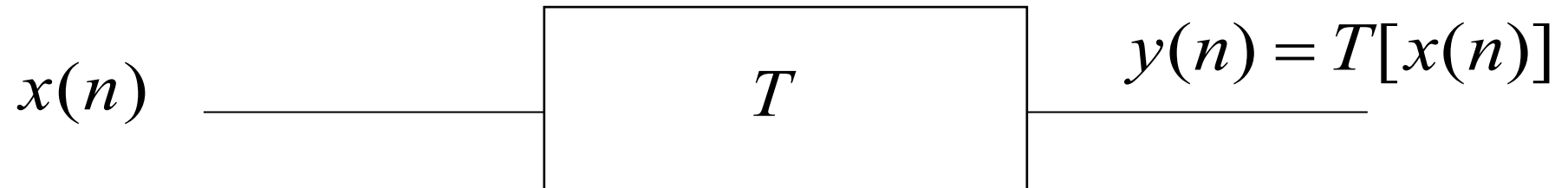
n	$u(n)$	$u(-2-n)$
-2	$u(-2)=0$	$u(-2+2)=1$
-1	$u(-1)=0$	$u(-2+1)=0$
0	$u(0)=1$	$u(-2-0)=0$
1	$u(1)=1$	$u(-2-1)=0$
2	$u(2)=1$	$u(-2-2)=0$
3	$u(3)=1$	$u(-2-3)=0$
4	$u(4)=1$	$u(-2-4)=0$

$$x(n+1)u(2-n)$$

n	x(n)	x(n+1)	u(n)	Result x(n+1)u(n)
-2	0	$x(-2+1)=x(-1)$	$u(2+2)=1$	0
-1	0	$x(-1+1) = x(0)=1$	$u(2+1)=1$	1
0	1	$x(0+1) = x(1)=1$	$u(2-0)=1$	1
1	1	$x(1+1) = x(2)=1$	$u(2-1)=1$	1
2	1	$x(2+1) = x(3)=.5$	$u(2-2)=1$	0.5
3	0.5	$x(3+1) = x(4)$	$u(2-3)=0$	0
4	0	$x(4+1)=x(5)$	$u(2-4)=0$	0
5	0	$X(5+1)=x(6)$	$u(2+5)=0$	0
6	0	$X(6+1)=x(7)$	$u(2+6)=0$	0

Linear Time-invariant (LTI) Systems

- A discrete-time system $T[]$ can either be
 - Linear or Non-linear
 - Time-invariant or Time-varying



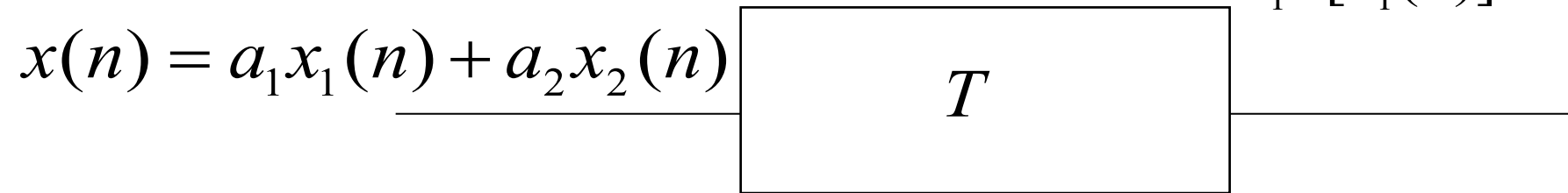
- We prefer systems that are **Linear and Time-invariant**.

Linear Systems

- For a system to be linear,

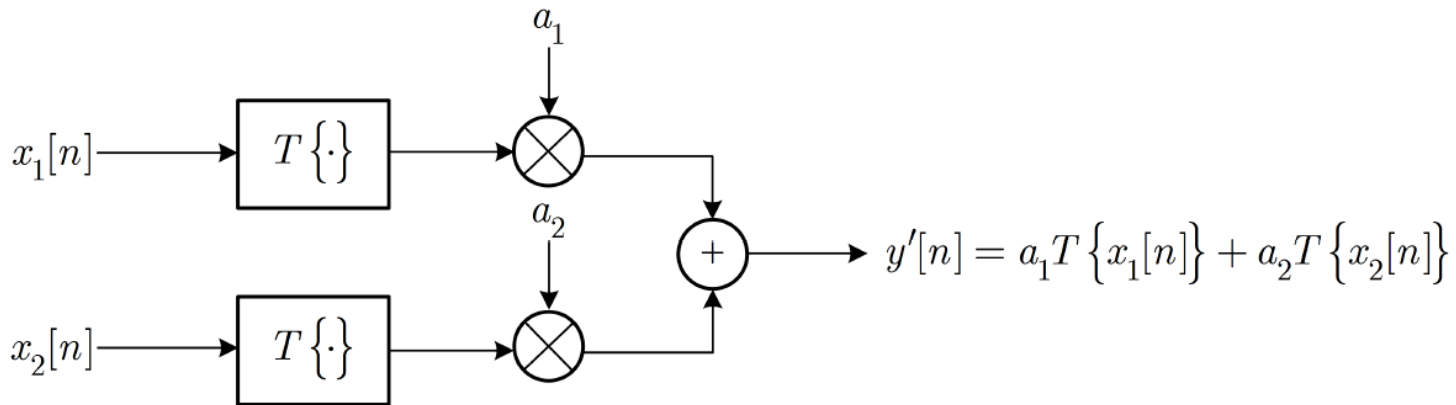
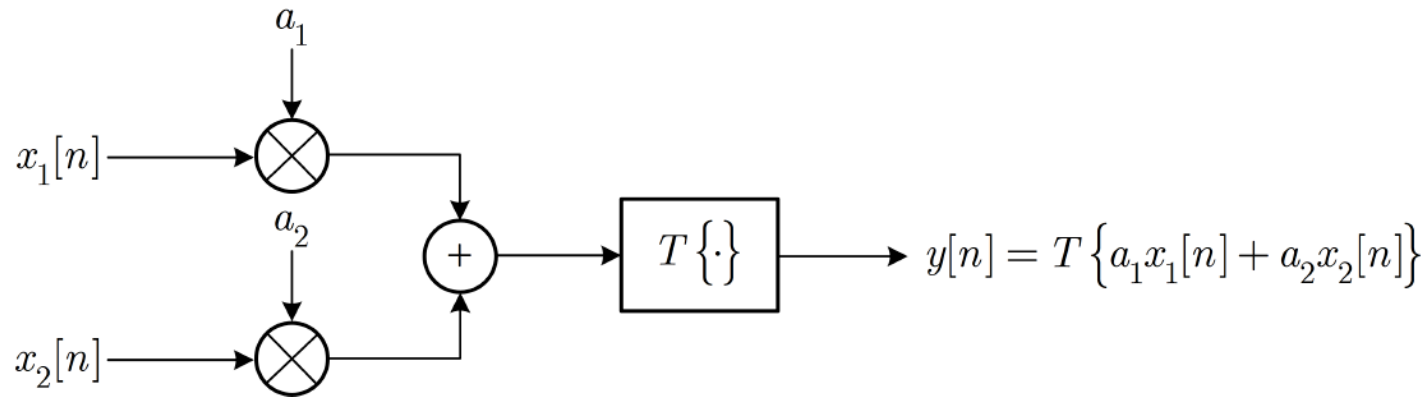
$$y(n) = a_1 y_1(n) + a_2 y_2(n)$$

$$= a_1 T[x_1(n)] + a_2 T[x_2(n)]$$



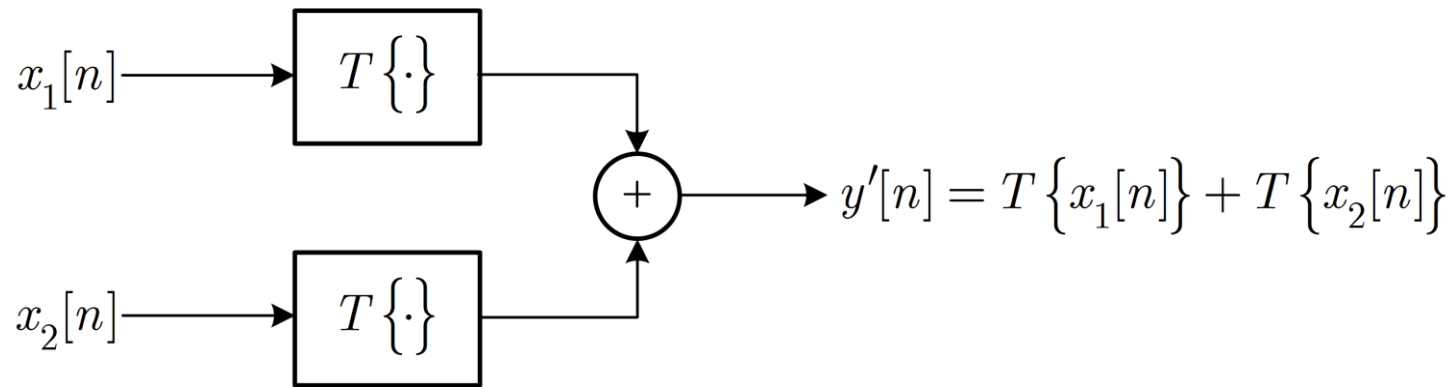
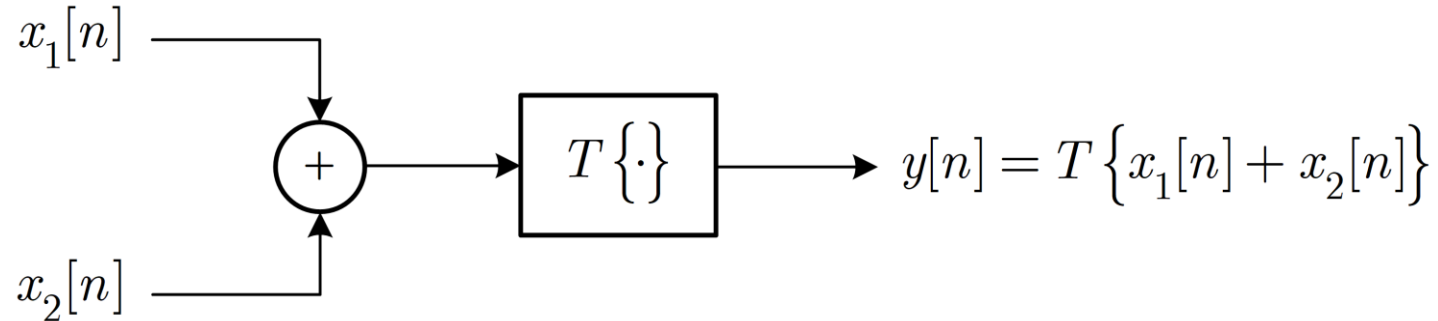
$$T[a_1 x_1(n) + a_2 x_2(n)] = a_1 T[x_1(n)] + a_2 T[x_2(n)]$$

How to test linearity: Full Test



Linear $T[a_1x_1(n) + a_2x_2(n)] = a_1T[x_1(n)] + a_2T[x_2(n)]$

How to test linearity: Simplified (Drop all constants)



Linear

$$T[x_1(n) + x_2(n)] = T[x_1(n)] + T[x_2(n)]$$

Example 2.2

- Determine whether the systems are linear or non-linear.

$$A) \quad y(n) = nx(n)$$

$$B) \quad y(n) = x(n^2)$$

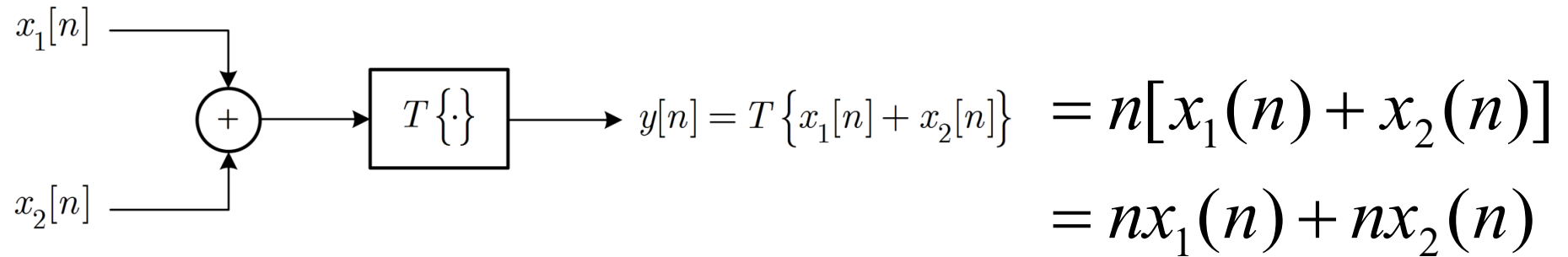
$$C) \quad y(n) = x^2(n)$$

$$D) \quad y(n) = Ax(n) + B$$

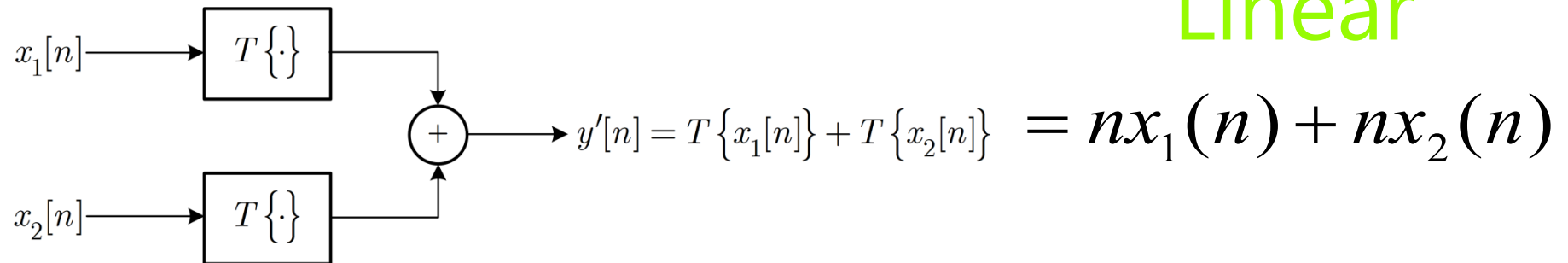
$$E) \quad y(n) = e^{x(n)}$$

Example 2.2: A)

A) $y(n) = nx(n)$

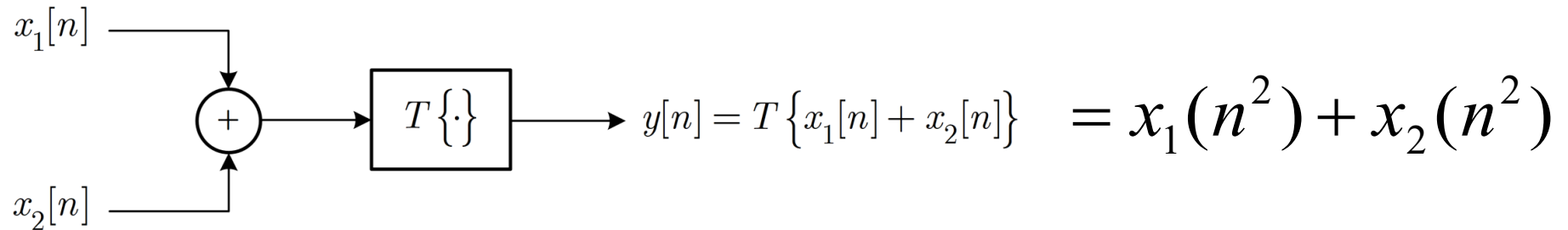


Linear

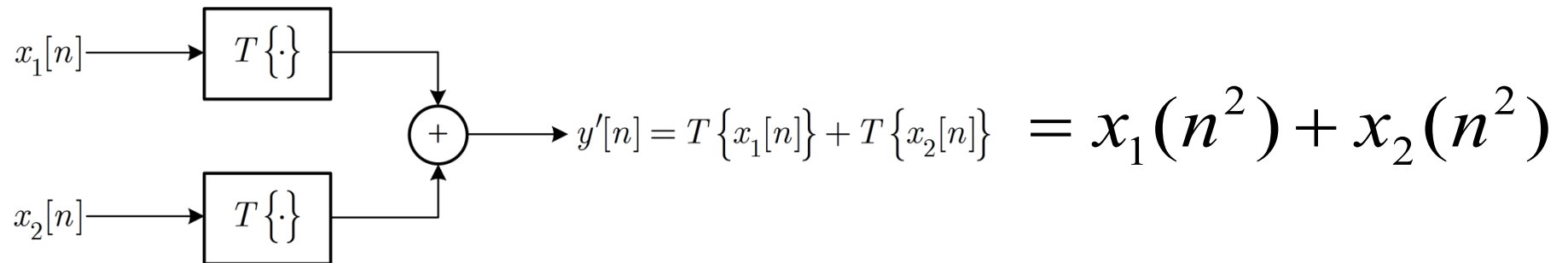


Example 2.2: B)

B) $y(n) = x(n^2)$

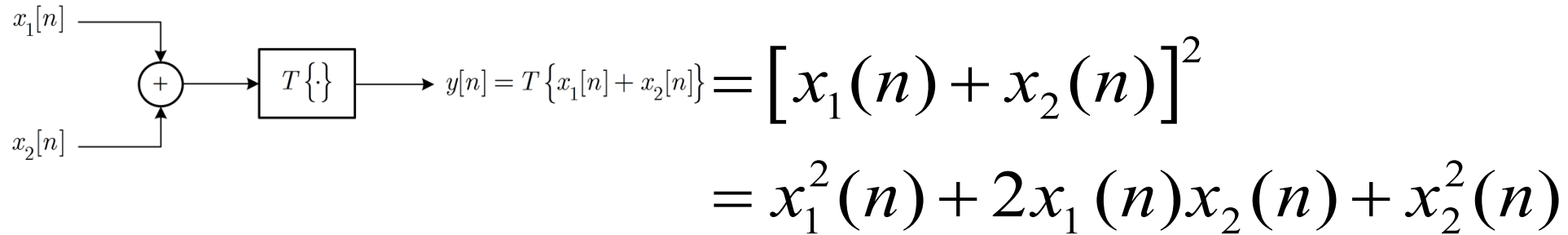


Linear

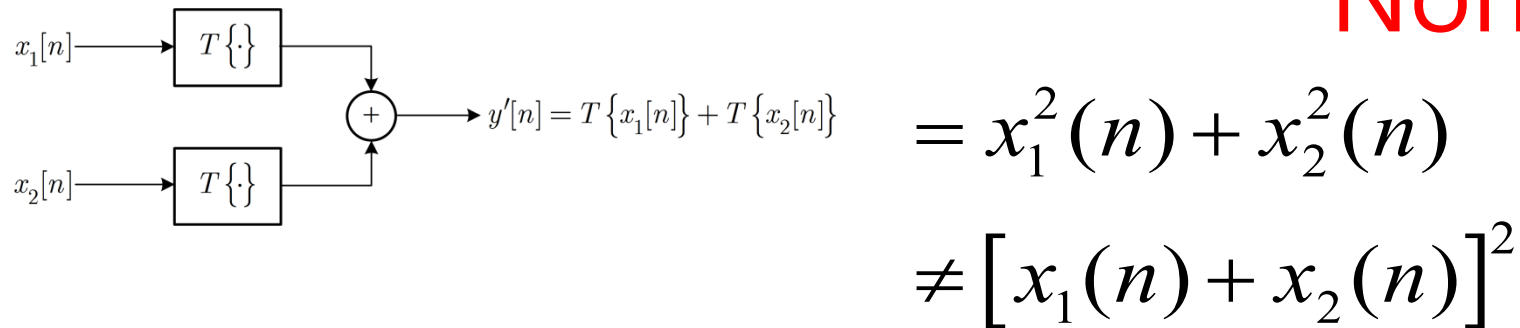


Example 2.2: C)

C) $y(n) = x^2(n)$

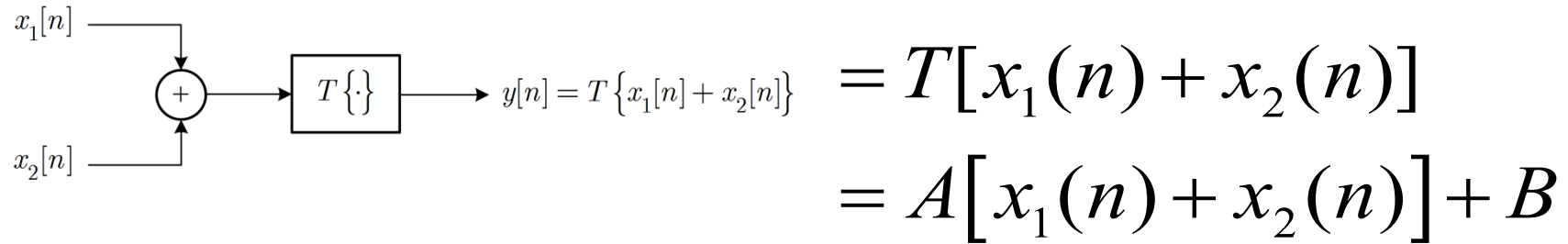


Non-Linear

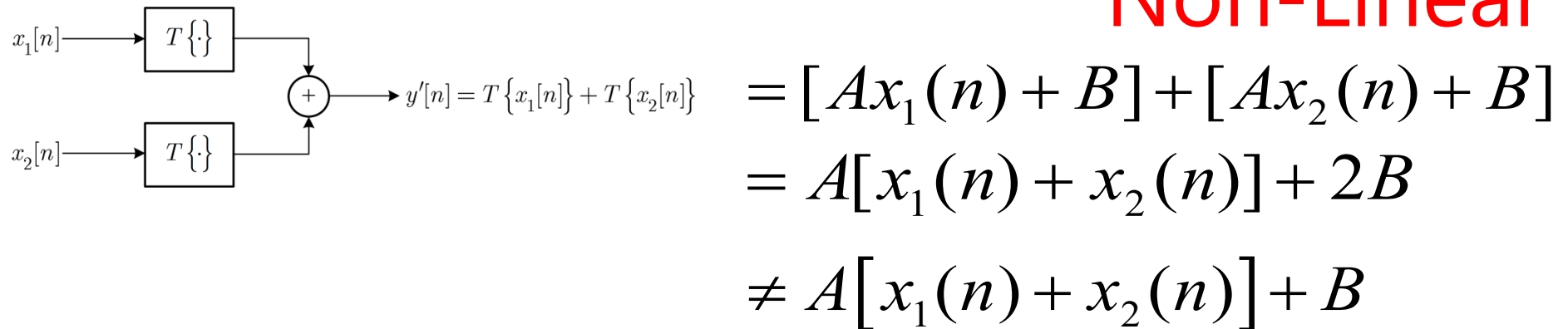


Example 2.2: D)

$$D) \quad y(n) = Ax(n) + B$$

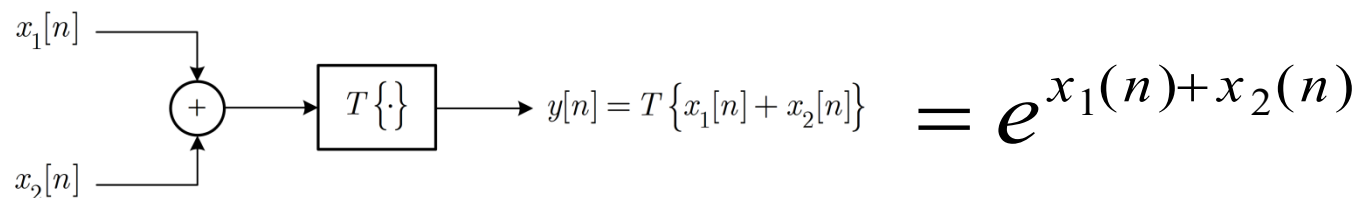


Non-Linear

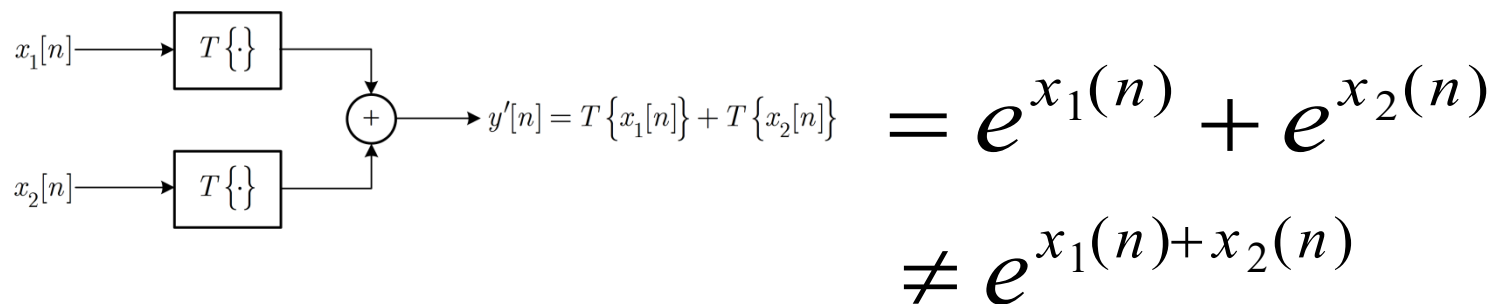


Example 2.2: E)

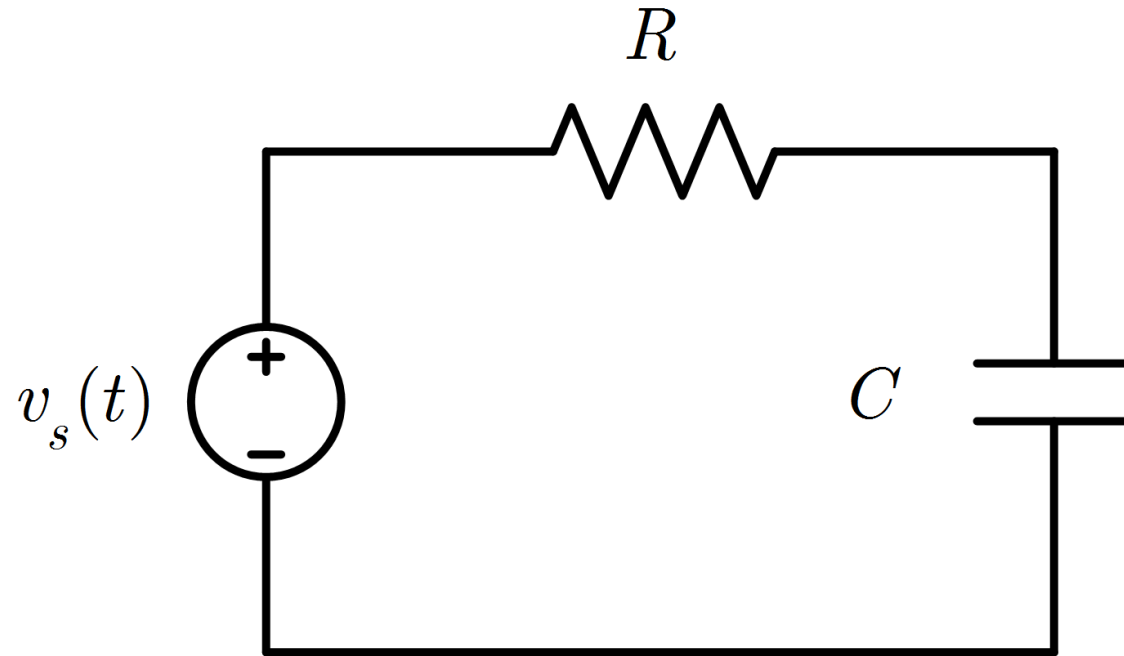
$$E) \quad y(n) = e^{x(n)}$$



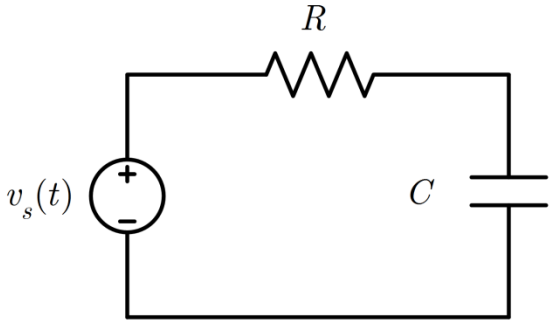
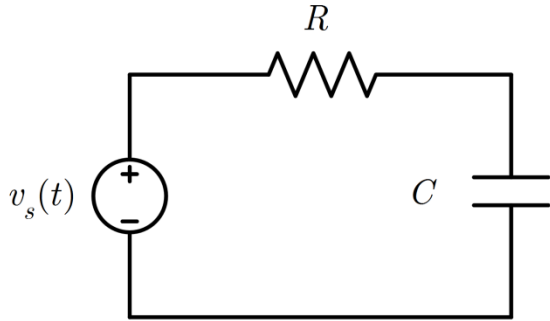
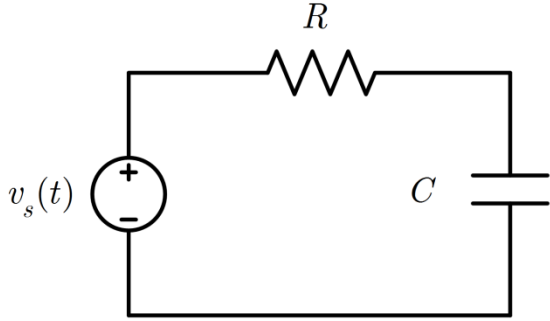
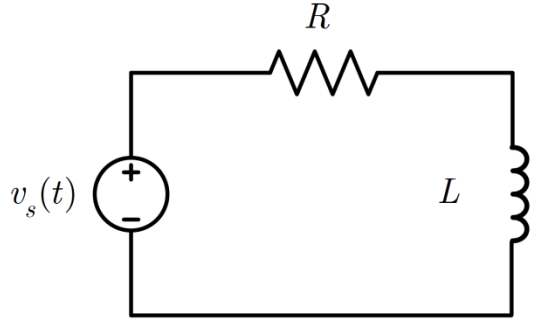
Non-Linear



Is this RC circuit Time-Invariant?

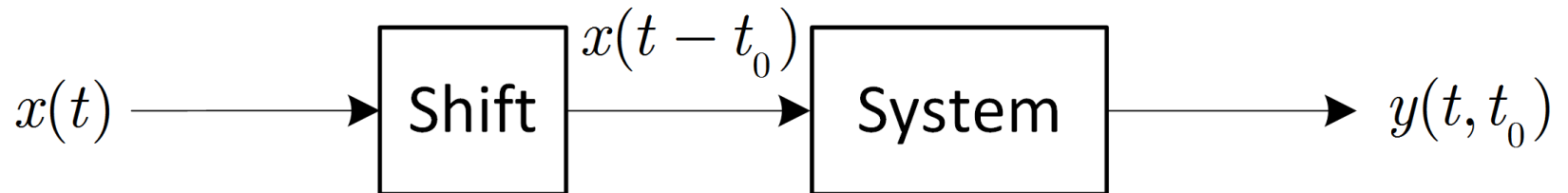
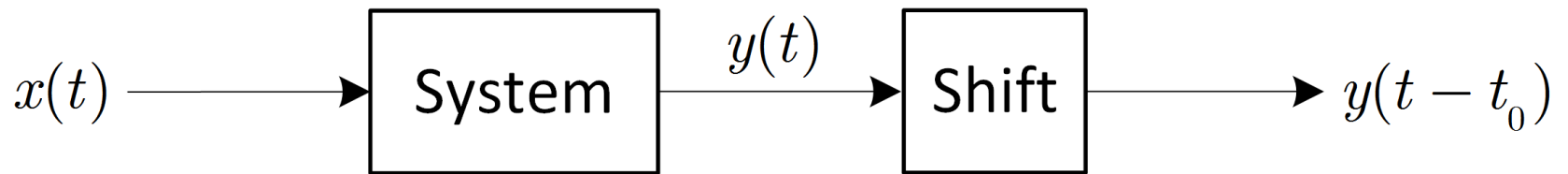


Time-invariant

t	Time Invariant	Time Varying
0	 A series circuit containing a voltage source $v_s(t)$, a resistor R, and a capacitor C.	 A series circuit containing a voltage source $v_s(t)$, a resistor R, and a capacitor C.
1000	 A series circuit containing a voltage source $v_s(t)$, a resistor R, and a capacitor C.	 A series circuit containing a voltage source $v_s(t)$, a resistor R, and an inductor L.

Time-Invariant Test (Continuous-Time Systems)

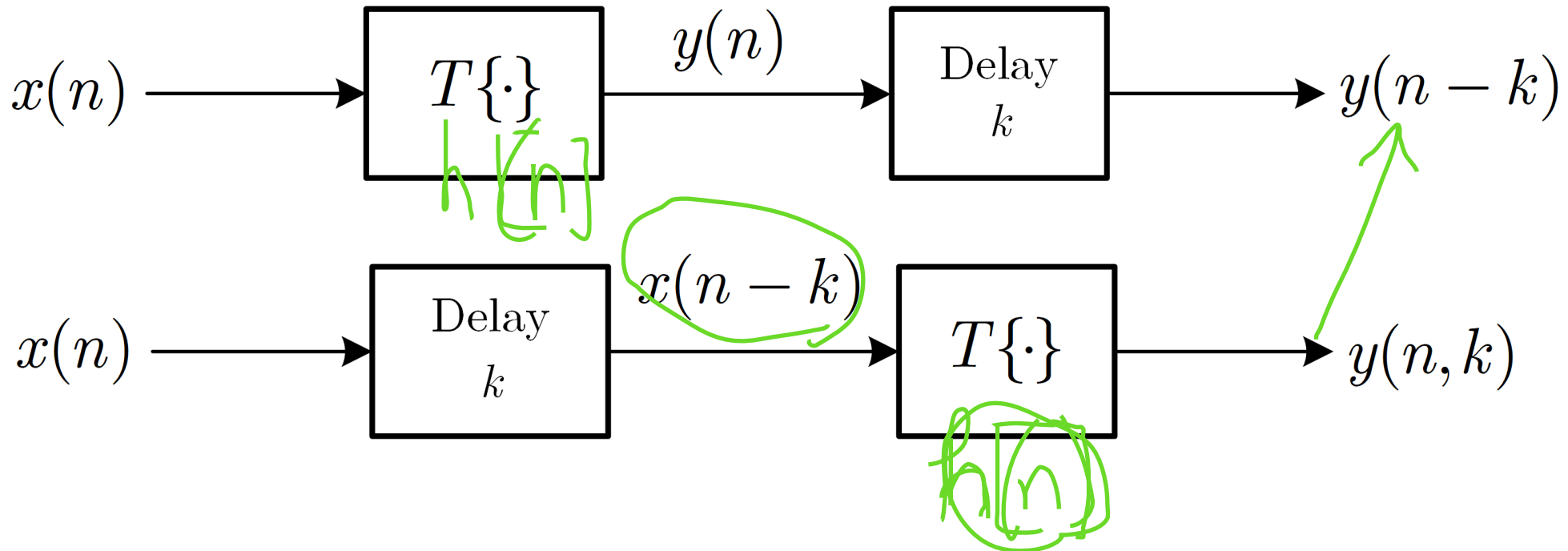
- Tested by comparing the following systems.



Time-Invariant

$$y(t, t_0) = y(t - t_0)$$

Time-Invariant Test (Discrete-Time Systems)



Time-Invariant

$$y(n,k) = y(n-k)$$

Example 2.3: Time-Invariant 1

- Determine whether the following systems are Time-invariant or Time-varying

$$A) \quad y(n) = x(n) - x(n-1)$$

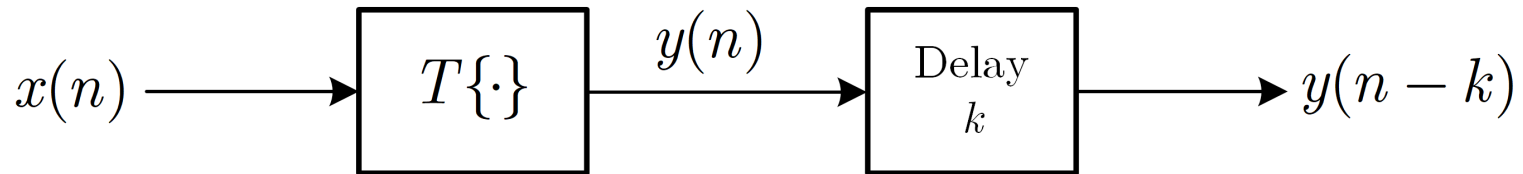
$$B) \quad y(n) = nx(n)$$

$$C) \quad y(n) = x(-n)$$

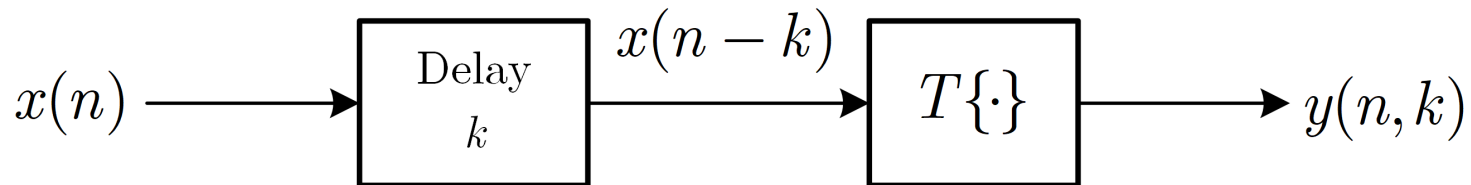
$$D) \quad y(n) = x(n) \cos \omega_0 n$$

Example 2.3: Time-Invariant A)

$$A) \quad y(n) = x(n) - x(n-1)$$



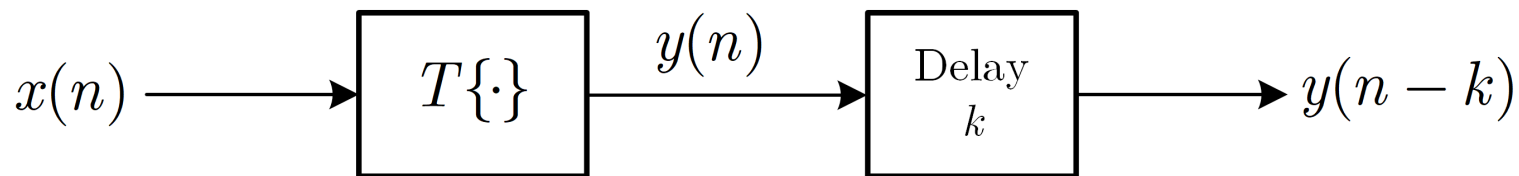
$$y(n-k) = x(n-k) - x(n-k-1)$$



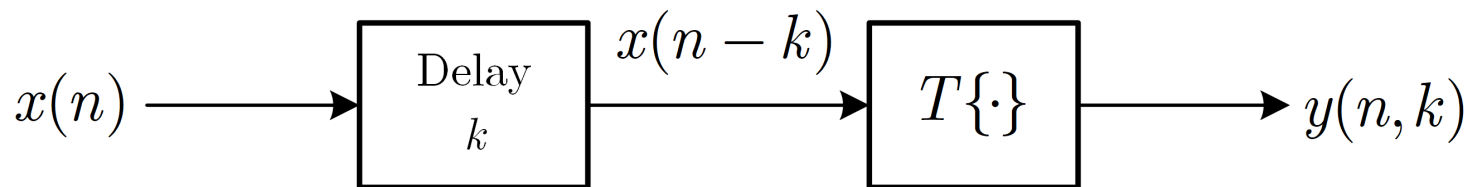
$$y(n, k) = T[x(n-k)] \quad \text{Time-invariant}$$
$$= x(n-k) - x(n-k-1)$$

Example 2.3: Time-Invariant B)

$$B) \quad y(n) = nx(n)$$



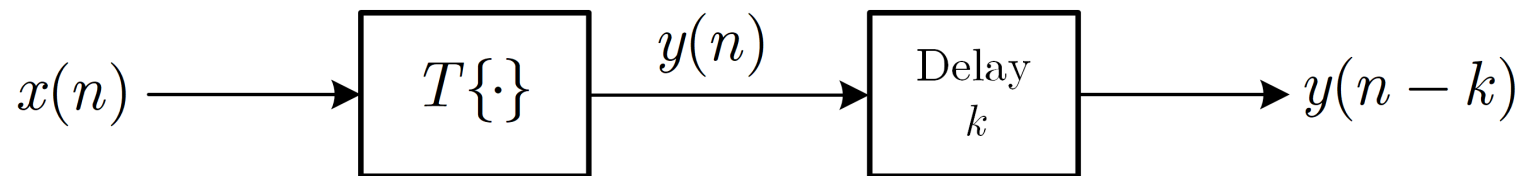
$$y(n-k) = (n-k)x(n-k)$$



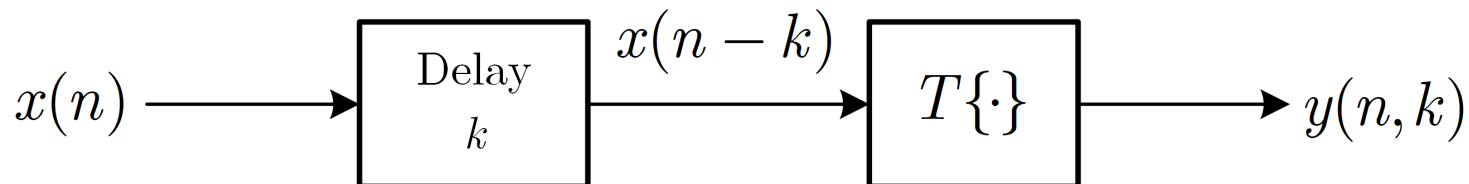
$$y(n,k) = T[x(n-k)] \text{ Time-varying} \\ = nx(n-k)$$

Example 2.3: Time-Invariant C)

$$C) \quad y(n) = x(-n)$$



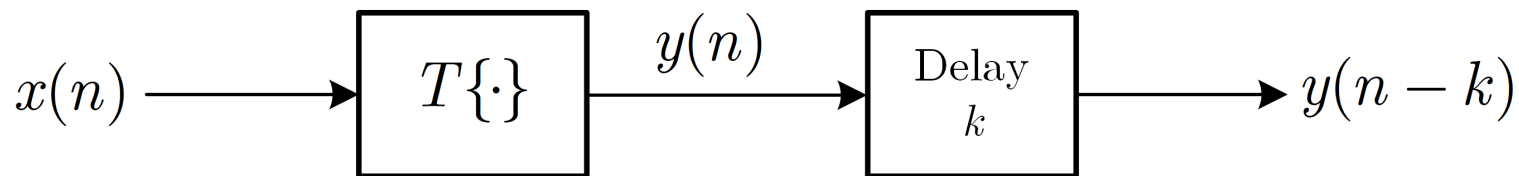
$$y(n - k) = x(-(n - k)) = x(-n + k)$$



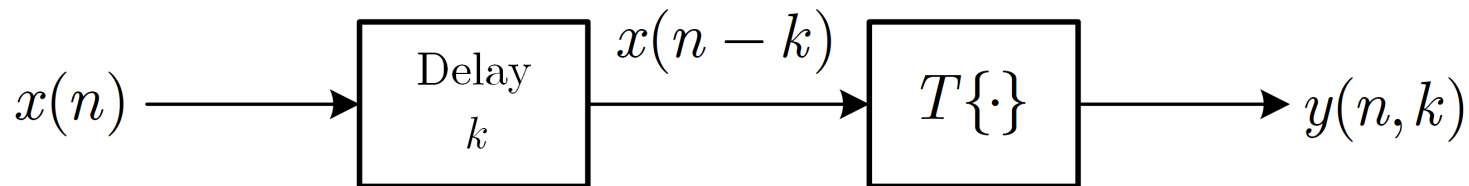
$$\begin{aligned} y(n, k) &= T[x(-n - k)] \text{Time-varying} \\ &= x(-n - k) \end{aligned}$$

Example 2.3: Time-Invariant D)

$$D) \quad y(n) = x(n) \cos \omega_0 n$$



$$y(n - k) = x(n - k) \cos \omega_0 (n - k)$$



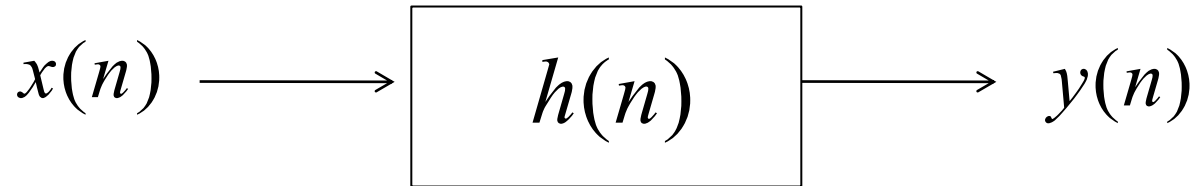
$$\begin{aligned} y(n, k) &= T[x(n) \cos \omega_0 n] \text{Time-varying} \\ &= x(n - k) \cos \omega_0 (n) \end{aligned}$$

Why is the LTI Systems so Important?

- The LTI property allows us to analyse the characteristics of the system.
- Above all, with this LTI property, the system can be used to find the relationship between the input, the impulse response and the output of the system.
- We are talking about the Convolution.

Convolution

- Convolution is a method to find output of the system under investigation.



$$y(n) = h(n) * x(n)$$

The Convolution related to the Delta function

- It involves the important property of the delta function: Sifting

An Impulse is Delta Function

$$\delta(n) = \delta(t) \Big|_{t=nT}$$

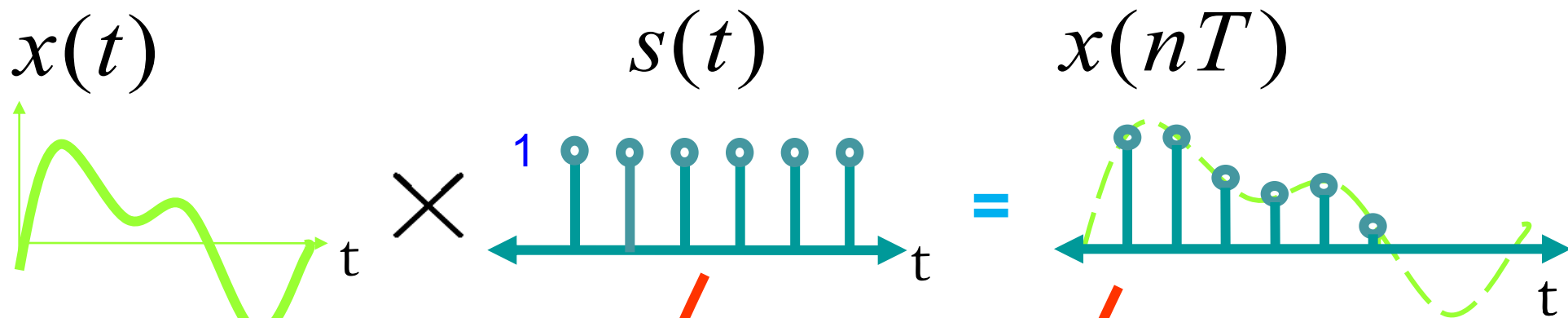


- The definition of impulse is

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Discrete-time Signal $x(n)$ from $x(t)$

- $x(n)$ is $x(t)$ with $t=nT$



$$x(t) \sum_{k=-\infty}^{\infty} \delta(t - kT) = x(nT)$$

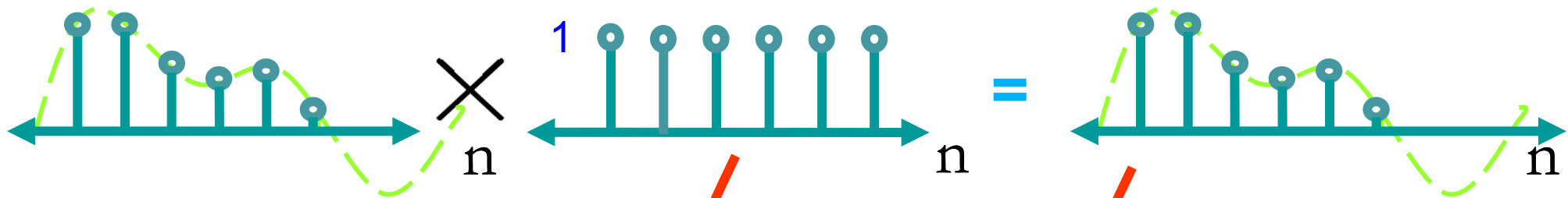
Discrete-time Signal $x(n)$ from $x(n)$

Previously, we know that

$x(n)$

$s(n)$

$x(n)$



$$\sum_{k=-\infty}^{\infty} x(k) \delta(n - k) = x(n)$$

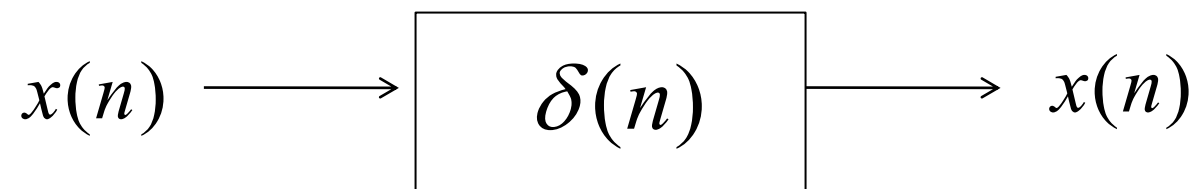
Sampling as a Convolution

- So the signal $x(n)$ which is constructed by sampling $x(n)$ **itself** is written as

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$$

- We can say it as the **Convolution** between $x(n)$ and $\delta(n)$

Sampling as a Convolution as System



$$x(n) = x(n) * \delta(n) = \sum_{k=-\infty}^{\infty} x(k) \delta(n - k)$$

Calculate Convolution at $n=0$

$n = 0$ and let $k = -1, 0, 1, 3$

$$x(n) = x(n) * \delta(n) = \sum_{k=-\infty}^{\infty} x(k) \delta(n - k)$$

$$x(-1)\delta(1) + x(0)\delta(0) + x(1)\delta(-1) + x(2)\delta(-2) + x(3)\delta(-3)$$

$$= x(0) \qquad \delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Calculate Convolution at $n=1$

$n = 1$ and let $k = -1, 0, 1, 3$

$$x(n) = x(n) * \delta(n) = \sum_{k=-\infty}^{\infty} x(k) \delta(n - k)$$

$$x(-1)\delta(2) + x(0)\delta(1) + \boxed{x(1)\delta(0)} + x(2)\delta(-1) + x(3)\delta(-2)$$

$$= x(1)$$
$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Calculate Convolution at $n=2$

$n = 2$ and let $k = -1, 0, 1, 3$

$$x(n) = x(n) * \delta(n) = \sum_{k=-\infty}^{\infty} x(k) \delta(n - k)$$

$$x(-1)\delta(3) + x(0)\delta(2) + x(1)\delta(1) + \boxed{x(2)\delta(0)} + x(3)\delta(-1)$$

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

$$= x(2)$$

Calculate Convolution at $n=3$

$n = 3$ and let $k = -1, 0, 1, 3$

$$x(n) = x(n) * \delta(n) = \sum_{k=-\infty}^{\infty} x(k) \delta(n - k)$$

$$x(-1)\delta(4) + x(0)\delta(3) + x(1)\delta(2) + x(2)\delta(1) + x(3)\delta(0)$$

$$= x(3)$$
$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Convolution of delta function and $x(n)$

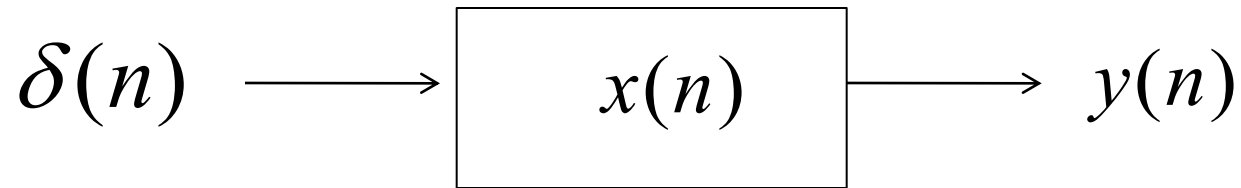
$$x(n) = \delta(n) * x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$$

n	$x(n) = \sum_{k=-1}^3 x(k)\delta(n-k)$
0	$x(-1)\delta(1) + \boxed{x(0)\delta(0)} + x(1)\delta(-1) + x(2)\delta(-2) + x(3)\delta(-3)$
1	$x(-1)\delta(2) + x(0)\delta(1) + \boxed{x(1)\delta(0)} + x(2)\delta(-1) + x(3)\delta(-2)$
2	$x(-1)\delta(3) + x(0)\delta(2) + x(1)\delta(1) + \boxed{x(2)\delta(0)} + x(3)\delta(-1)$
3	$x(-1)\delta(4) + x(0)\delta(3) + x(1)\delta(2) + x(2)\delta(1) + \boxed{x(3)\delta(0)}$

n	$x(n) = \sum_{k=-2}^2 x(k)\delta(n-k)$
0	$x(0)$
1	$x(1)$
2	$x(2)$
3	$x(3)$

$$x(n) = x(n)$$

Convolution of Delta function with $x(n)$ as the impulse response



$$y(n) = \delta(n) * x(n) = \sum_{k=-\infty}^{\infty} \delta(k)x(n-k)$$

Calculate Convolution at $n=0$

$n = 0$ and let $k = -2, -1, 0, 1, 2$

$$y(n) = \delta(n) * x(n) = \sum_{k=-2}^2 \delta(k)x(n-k)$$

$$= \delta(-2)x(2) + \delta(-1)x(1) + \delta(0)x(0) + \delta(1)x(-1) + \delta(2)x(-2)$$

$$= x(0)$$

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Calculate Convolution at $n=1$

$n = 1$ and let $k = -2, -1, 0, 1, 2$

$$y(n) = \delta(n) * x(n) = \sum_{k=-2}^2 \delta(k)x(n-k)$$

$$\delta(-2)x(3) + \delta(-1)x(2) + \delta(0)x(1) + \delta(1)x(0) + \delta(2)x(1)$$

$$= x(1)$$

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Calculate Convolution at $n=2$

$n = 2$ and let $k = -2, -1, 0, 1, 2$

$$y(n) = \delta(n) * x(n) = \sum_{k=-2}^2 \delta(k)x(n-k)$$

$$\delta(-2)x(4) + \delta(-1)x(3) + \delta(0)x(2) + \delta(1)x(1) + \delta(2)x(0)$$

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

$$= x(2)$$

Calculate Convolution at $n=3$

$n = 3$ and let $k = -2, -1, 0, 1, 2$

$$y(n) = \delta(n) * x(n) = \sum_{k=-2}^2 \delta(k)x(n-k)$$

$$\delta(-2)x(5) + \delta(-1)x(4) + \delta(0)x(3) + \delta(1)x(2) + \delta(2)x(1)$$

$$= x(3)$$

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Convolution of $x(n)$ and delta function

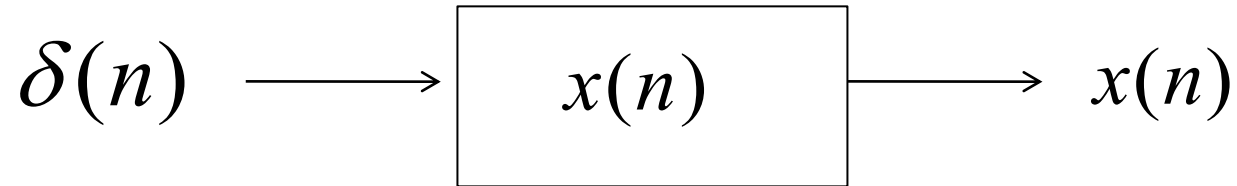
$$y(n) = \delta(n) * x(n) = \sum_{k=-\infty}^{\infty} \delta(k)x(n-k)$$

n	$y(n) = \sum_{k=-2}^2 \delta(k)x(n-k)$
0	$\delta(-2)x(2) + \delta(-1)x(1) + \boxed{\delta(0)x(0)} + \delta(1)x(-1) + \delta(2)x(-2)$
1	$\delta(-2)x(3) + \delta(-1)x(2) + \boxed{\delta(0)x(1)} + \delta(1)x(0) + \delta(2)x(1)$
2	$\delta(-2)x(4) + \delta(-1)x(3) + \boxed{\delta(0)x(2)} + \delta(1)x(1) + \delta(2)x(0)$
3	$\delta(-2)x(5) + \delta(-1)x(4) + \boxed{\delta(0)x(3)} + \delta(1)x(2) + \delta(2)x(1)$

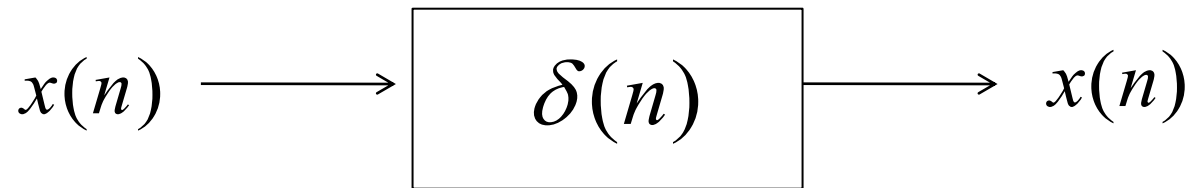
n	$y(n) = \sum_{k=-2}^2 \delta(k)x(n-k)$
0	$x(0)$
1	$x(1)$
2	$x(2)$
3	$x(3)$

$$y(n) = x(n)$$

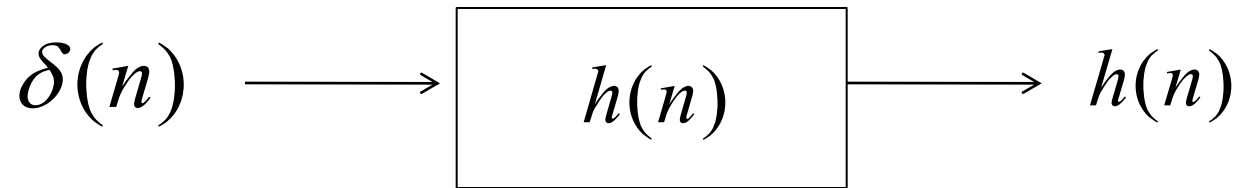
The input $x(n)$ and delta function are interchangeable



$$x(n) = \delta(n) * x(n) = \sum_{k=-\infty}^{\infty} \delta(k)x(n-k)$$



$$x(n) = x(n) * \delta(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$$



$h(n)$ = Impulse Response

$$h(n) = \delta(n) * h(n) = \sum_{k=-\infty}^{\infty} \delta(k)h(n-k)$$

Comparison between the two convolutions

Convolution of $x(n)$ and delta function

$$x(n) = \delta(n) * x(n) = \sum_{k=-\infty}^{\infty} \delta(k)x(n-k)$$

n	$x(n) = \sum_{k=-2}^2 \delta(k)x(n-k)$
0	$\delta(-2)x(2) + \delta(-1)x(1) + \boxed{\delta(0)x(0)} + \delta(1)x(-1) + \delta(2)x(-2)$
1	$\delta(-2)x(3) + \delta(-1)x(2) + \boxed{\delta(0)x(1)} + \delta(1)x(0) + \delta(2)x(1)$
2	$\delta(-2)x(4) + \delta(-1)x(3) + \boxed{\delta(0)x(2)} + \delta(1)x(1) + \delta(2)x(0)$
3	$\delta(-2)x(5) + \delta(-1)x(4) + \boxed{\delta(0)x(3)} + \delta(1)x(2) + \delta(2)x(1)$

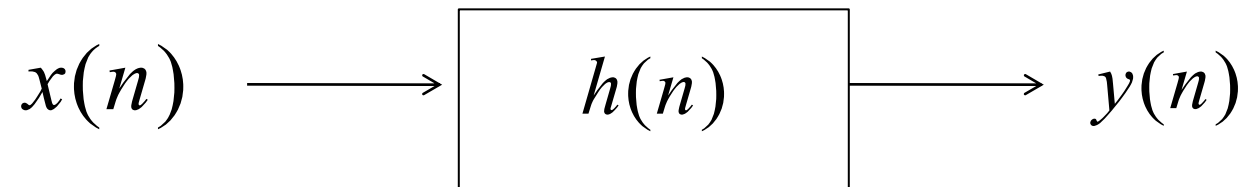
Convolution of delta function and $x(n)$

$$x(n) = \delta(n) * x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$$

n	$x(n) = \sum_{k=-1}^3 x(k)\delta(n-k)$
0	$x(-1)\delta(1) + \boxed{x(0)\delta(0)} + x(1)\delta(-1) + x(2)\delta(-2) + x(3)\delta(-3)$
1	$x(-1)\delta(2) + x(0)\delta(1) + \boxed{x(1)\delta(0)} + x(2)\delta(-1) + x(3)\delta(-2)$
2	$x(-1)\delta(3) + x(0)\delta(2) + x(1)\delta(1) + \boxed{x(2)\delta(0)} + x(3)\delta(-1)$
3	$x(-1)\delta(4) + x(0)\delta(3) + x(1)\delta(2) + x(2)\delta(1) + \boxed{x(3)\delta(0)}$

Convolution in the General Form

- $x(n)$ is the input $h(n)$ the impulse response and $y(n)$ the output

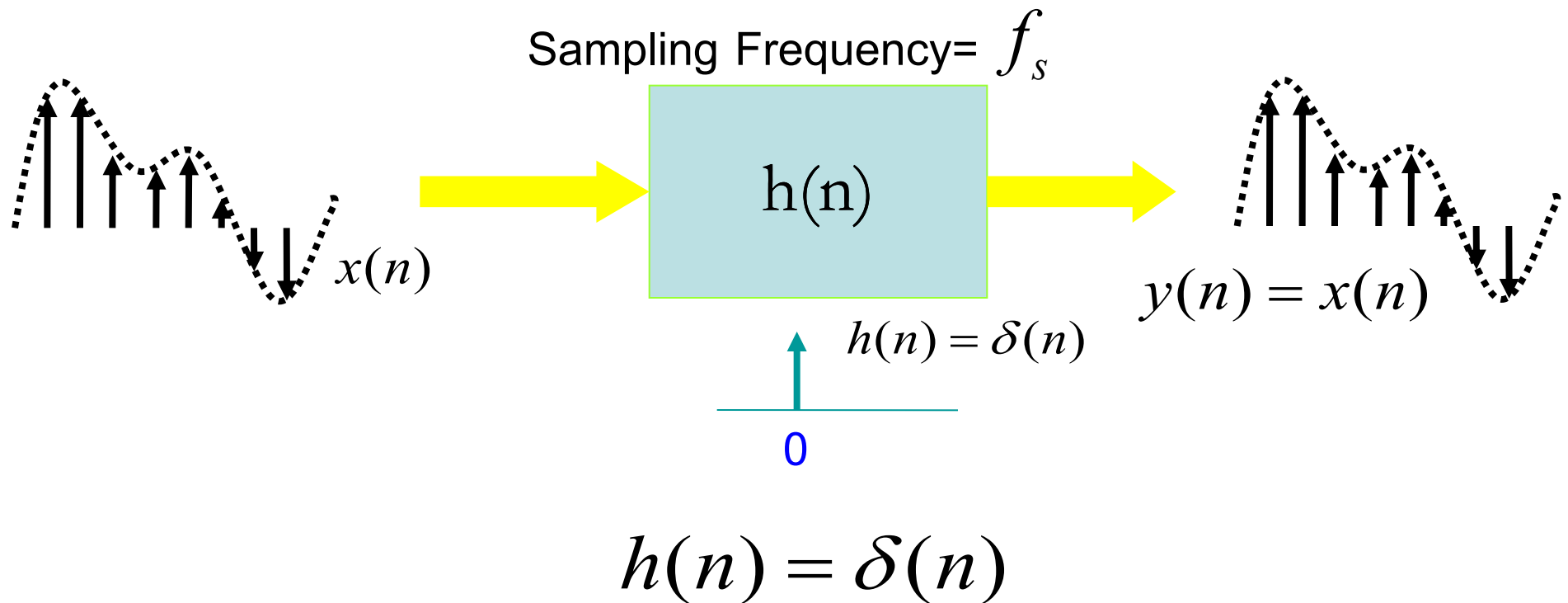


$$y(n) = h(n) * x(n) = x(n) * h(n)$$

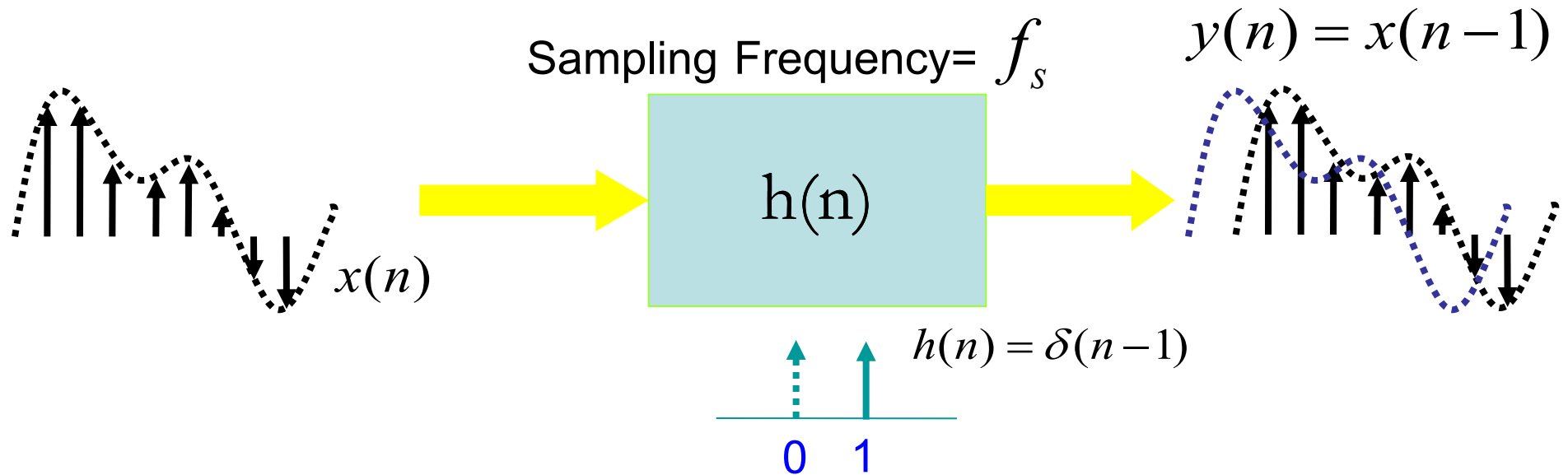
$$= \sum_{k=-\infty}^{\infty} h(k)x(n-k) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

Convolution can be used to determine the output signal of a LTI system for a given input signal with knowledge of the system's unit impulse response.

System with the Delta function as the Impulse Response



System with a Unit Delayed Delta function as the Impulse Response



$$h(n] = \delta(n-1]$$

$$y(n) = \sum_{k=-2}^2 \delta(k-1)x(n-k)$$

$$= \delta(-3)x(n+2) + \delta(-2)x(n+1) + \delta(-1)x(n) + \boxed{\delta(0)x(n-1)} + \delta(1)x(-2)$$

$$y(n) = \sum_{k=-2}^2 \delta(k-1)x(n-k)$$

n	$y(n) = \sum_{k=-2}^2 \delta(k-1)x(n-k)$
0	$\delta(-3)x(2) + \delta(-2)x(1) + \delta(-1)x(0) + \boxed{\delta(0)x(-1)} + \delta(1)x(-2)$
1	$\delta(-3)x(3) + \delta(-2)x(2) + \delta(-1)x(1) + \boxed{\delta(0)x(0)} + \delta(1)x(-1)$
2	$\delta(-3)x(4) + \delta(-2)x(3) + \delta(-1)x(2) + \boxed{\delta(0)x(1)} + \delta(1)x(0)$
3	$\delta(-3)x(5) + \delta(-2)x(4) + \delta(-1)x(3) + \boxed{\delta(0)x(2)} + \delta(1)x(1)$

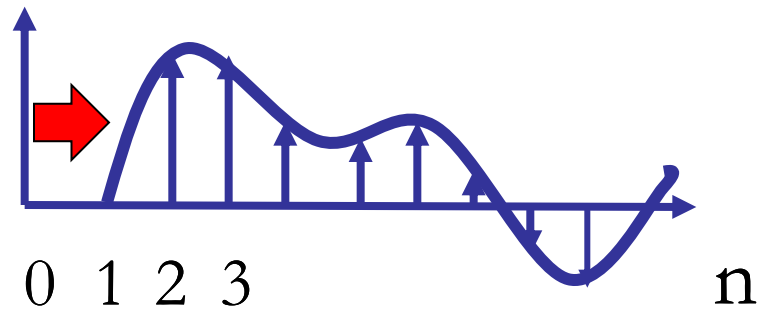
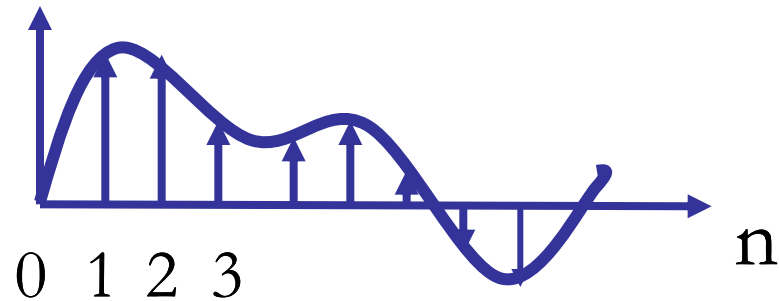
$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

n	$y(n) = \sum_{k=-2}^2 \delta(k)x(n-k)$
0	$x(-1)$
1	$x(0)$
2	$x(1)$
3	$x(2)$

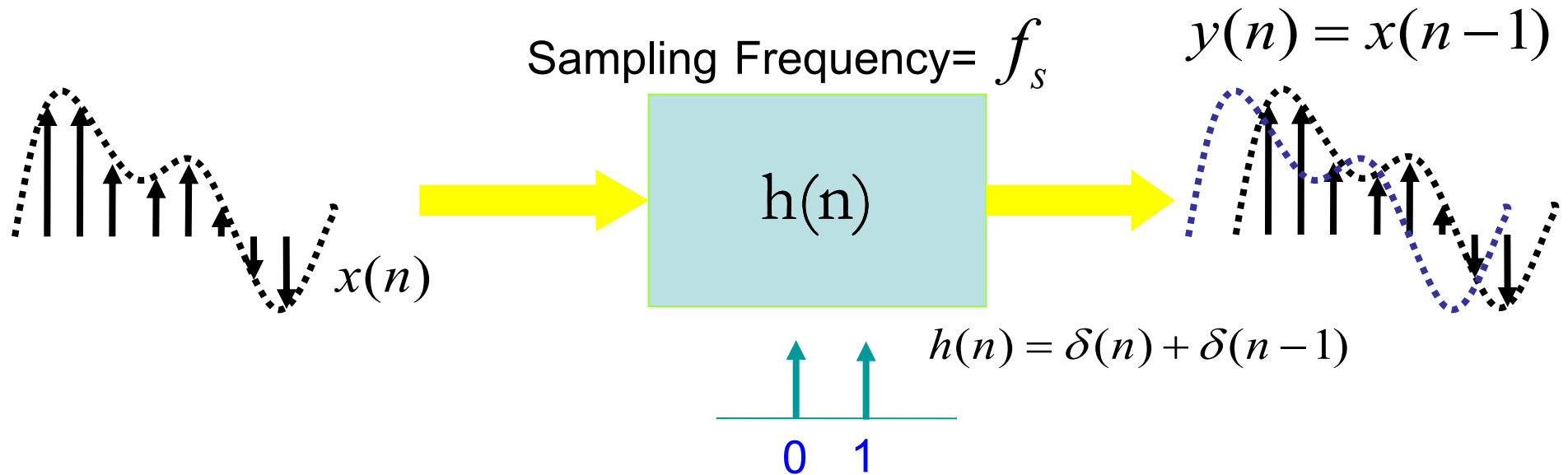
$$y(n) = x(n-1)$$

Delayed Signal

$$y(n] = x(n - 1)$$



System with and without Delayed Delta functions



$$h(n) = \delta(n) + \delta(n-1)$$

Convolution with System with and without Delayed Delta functions

- For any n ,

$$\begin{aligned}y(n) &= \sum_{k=-2}^2 [\delta(k) + \delta(k-1)]x(n-k) \\&= \sum_{k=-2}^2 \delta(k)x(n-k) + \sum_{k=-2}^2 \delta(k-1)x(n-k) \\&= \delta(-2)x(n+2) + \delta(-1)x(n+1) + \boxed{\delta(0)x(n)} + \delta(1)x(n-1) + \delta(2)x(-2) + \\&\quad \delta(-3)x(n+2) + \delta(-2)x(n+1) + \delta(-1)x(n) + \boxed{\delta(0)x(n-1)} + \delta(1)x(-2)\end{aligned}$$

Calculate Convolution at $n=0$

$n = 0$ and let $k = -2, -1, 0, 1, 2$

$$\begin{aligned} y(0) &= \sum_{k=-2}^2 \delta(k)x(n-k) + \sum_{k=-2}^2 \delta(k-1)x(n-k) \\ &= \delta(-2)x(2) + \delta(-1)x(1) + \boxed{\delta(0)x(0)} + \delta(1)x(-1) + \delta(2)x(-2) + \\ &\quad \delta(-3)x(2) + \delta(-2)x(1) + \delta(-1)x(0) + \boxed{\delta(0)x(-1)} + \delta(1)x(-2) \\ &= x(0) + x(-1) \end{aligned}$$

Calculate Convolution at $n=1$

$n = 1$ and let $k = -2, -1, 0, 1, 2$

$$\begin{aligned} y(1) &= \sum_{k=-2}^2 \delta(k)x(n-k) + \sum_{k=-2}^2 \delta(k-1)x(n-k) \\ &= \delta(-2)x(3) + \delta(-1)x(2) + \boxed{\delta(0)x(1)} + \delta(1)x(0) + \delta(2)x(1) + \\ &\quad \delta(-3)x(3) + \delta(-2)x(2) + \delta(-1)x(1) + \boxed{\delta(0)x(0)} + \delta(1)x(-1) \\ &= x(1) + x(0) \end{aligned}$$

Calculate Convolution at $n=2$

$n = 2$ and let $k = -2, -1, 0, 1, 2$

$$\begin{aligned} y(2) &= \sum_{k=-2}^2 \delta(k)x(n-k) + \sum_{k=-2}^2 \delta(k-1)x(n-k) \\ &= \delta(-2)x(4) + \delta(-1)x(3) + \boxed{\delta(0)x(2)} + \delta(1)x(1) + \delta(2)x(0) + \\ &\quad \delta(-3)x(4) + \delta(-2)x(3) + \delta(-1)x(2) + \boxed{\delta(0)x(1)} + \delta(1)x(0) \\ &= x(2) + x(1) \end{aligned}$$

Calculate Convolution at $n=3$

$n = 3$ and let $k = -2, -1, 0, 1, 2$

$$\begin{aligned} y(3) &= \sum_{k=-2}^2 \delta(k)x(n-k) + \sum_{k=-2}^2 \delta(k-1)x(n-k) \\ &= \delta(-2)x(5) + \delta(-1)x(4) + \boxed{\delta(0)x(3)} + \delta(1)x(2) + \delta(2)x(1) + \\ &\quad \delta(-3)x(5) + \delta(-2)x(4) + \delta(-1)x(3) + \boxed{\delta(0)x(2)} + \delta(1)x(1) \\ &= x(3) + x(2) \end{aligned}$$

$$y(n) = \sum_{k=-2}^2 \delta(k-1)x(n-k)$$

n	$y(n) = \sum_{k=-2}^2 [\delta(k) + \delta(k-1)]x(n-k)$
0	$\delta(-2)x(2) + \delta(-1)x(1) + \boxed{\delta(0)x(0)} + \delta(1)x(-1) + \delta(2)x(-2) +$ $\delta(-3)x(2) + \delta(-2)x(1) + \delta(-1)x(0) + \boxed{\delta(0)x(-1)} + \delta(1)x(-2)$
1	$\delta(-2)x(3) + \delta(-1)x(2) + \boxed{\delta(0)x(1)} + \delta(1)x(0) + \delta(2)x(1) +$ $\delta(-3)x(3) + \delta(-2)x(2) + \delta(-1)x(1) + \boxed{\delta(0)x(0)} + \delta(1)x(-1)$
2	$\delta(-2)x(4) + \delta(-1)x(3) + \boxed{\delta(0)x(2)} + \delta(1)x(1) + \delta(2)x(0) +$ $\delta(-3)x(4) + \delta(-2)x(3) + \delta(-1)x(2) + \boxed{\delta(0)x(1)} + \delta(1)x(0)$
3	$\delta(-2)x(5) + \delta(-1)x(4) + \boxed{\delta(0)x(3)} + \delta(1)x(2) + \delta(2)x(1) +$ $\delta(-3)x(5) + \delta(-2)x(4) + \delta(-1)x(3) + \boxed{\delta(0)x(2)} + \delta(1)x(1)$

n	$y(n)$
0	$x(0) + x(-1)$
1	$x(1) + x(0)$
2	$x(2) + x(1)$
3	$x(3) + x(2)$

$$y(n) = x(n) + x(n - 1)$$

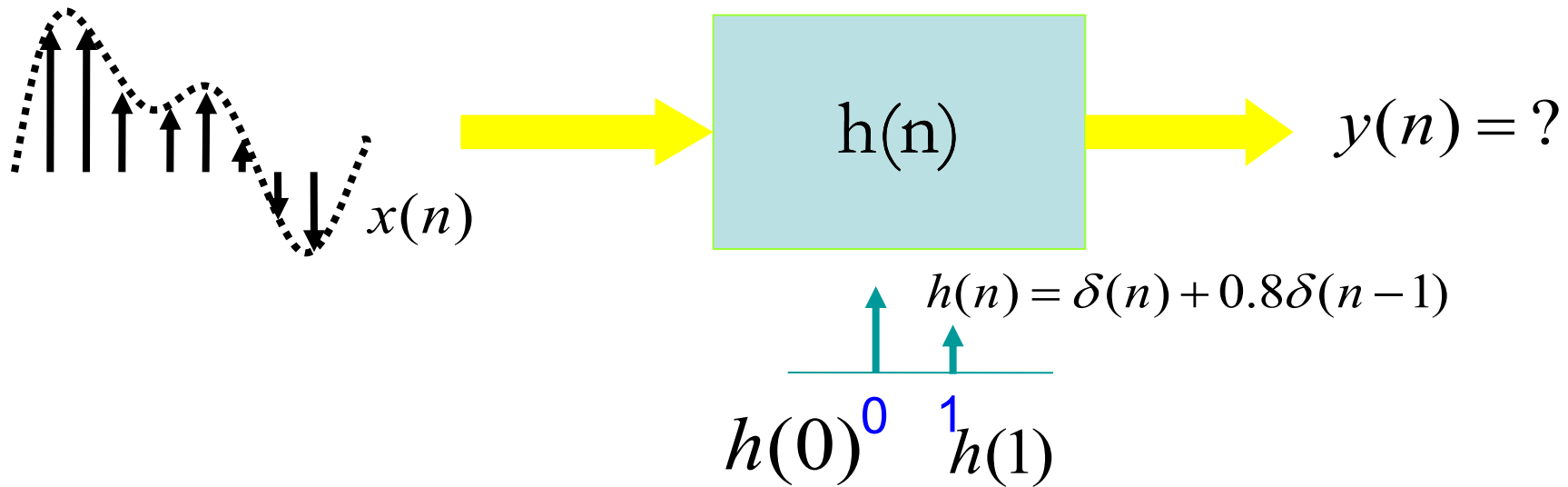
- Now if the impulse response is consisted of a scaled unit delayed

$$h(n) = \{\underset{\uparrow}{1}, 0.8\}$$

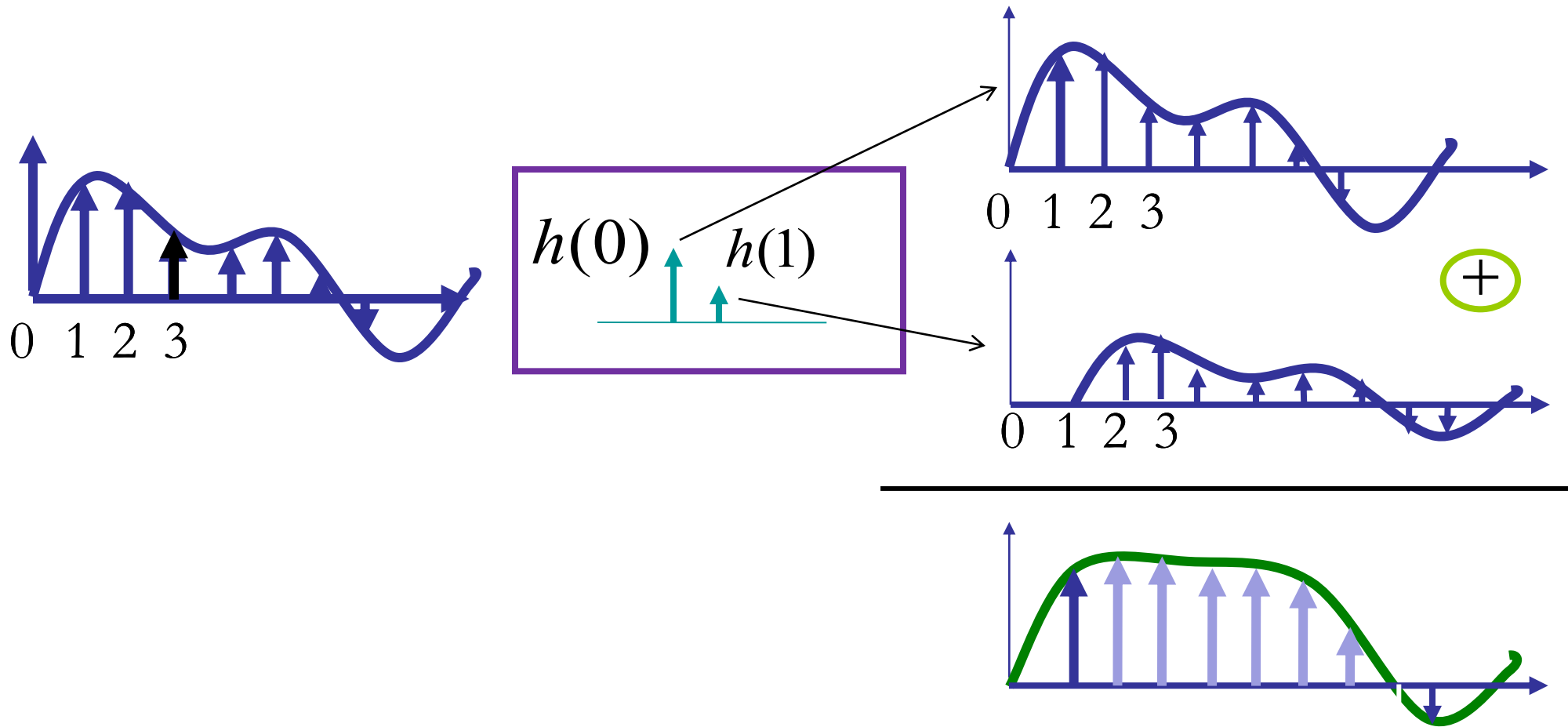
- Or $h(0) = 1$ and $h(1) = 0.8$
- This means we can write $h(n)$ as

$$h(n) = \delta(n) + 0.8\delta(n - 1)$$

Convolution of $x(n]$ and $h(n) = \{1, 0.8\}$



Convolved Signal



Calculate Convolution at $n=0$

$n = 0$ and let $k = -2, -1, 0, 1, 2$

$$\begin{aligned} y(0) &= \sum_{k=-2}^2 \delta(k)x(0-k) + 0.8 \sum_{k=-2}^2 \delta(k-1)x(0-k) \\ &= \delta(-2)x(2) + \delta(-1)x(1) + \boxed{\delta(0)x(0)} + \delta(1)x(-1) + \delta(2)x(-2) + \\ &\quad 0.8[\delta(-3)x(2) + \delta(-2)x(1) + \delta(-1)x(0) + \boxed{\delta(0)x(-1)} + \delta(1)x(-2)] \\ &= x(0) + 0.8x(-1) \end{aligned}$$

Calculate Convolution at n=1

n = 1 and let k=-2,-1,0,1,2

$$\begin{aligned} y(1) &= \sum_{k=-2}^2 \delta(k)x(1-k) + 0.8 \sum_{k=-2}^2 \delta(k-1)x(1-k) \\ &= \delta(-2)x(3) + \delta(-1)x(2) + \boxed{\delta(0)x(1)} + \delta(1)x(0) + \delta(2)x(1) + \\ &\quad 0.8 \left[\delta(-3)x(3) + \delta(-2)x(2) + \delta(-1)x(1) + \boxed{\delta(0)x(0)} + \delta(1)x(-1) \right] \\ &= x(1) + 0.8x(0) \end{aligned}$$

Calculate Convolution at n=2

n = 2 and let $k = -2, -1, 0, 1, 2$

$$\begin{aligned} y(2) &= \sum_{k=-2}^2 \delta(k)x(2-k) + 0.8 \sum_{k=-2}^2 \delta(k-1)x(2-k) \\ &= \delta(-2)x(4) + \delta(-1)x(3) + \boxed{\delta(0)x(2)} + \delta(1)x(1) + \delta(2)x(0) + \\ &\quad 0.8 \left[\delta(-3)x(4) + \delta(-2)x(3) + \delta(-1)x(2) + \boxed{\delta(0)x(1)} + \delta(1)x(0) \right] \\ &= x(2) + 0.8x(1) \end{aligned}$$

Calculate Convolution at $n=3$

$n = 3$ and let $k = -2, -1, 0, 1, 2$

$$\begin{aligned} y(3) &= \sum_{k=-2}^2 \delta(k)x(3-k) + 0.8 \sum_{k=-2}^2 \delta(k-1)x(3-k) \\ &= \delta(-2)x(5) + \delta(-1)x(4) + \boxed{\delta(0)x(3)} + \delta(1)x(2) + \delta(2)x(1) + \\ &\quad 0.8 \left[\delta(-3)x(5) + \delta(-2)x(4) + \delta(-1)x(3) + \boxed{\delta(0)x(2)} + \delta(1)x(1) \right] \\ &= x(3) + 0.8x(2) \end{aligned}$$

n	$y(n)$
0	$x(0) + 0.8x(-1)$
1	$x(1) + 0.8x(0)$
2	$x(2) + 0.8x(1)$
3	$x(3) + 0.8x(2)$

$$y(n) = x(n) + 0.8x(n-1)$$

Example 1: Convolution

- Determine the convolution results of $y(n)=x(n)*h(n)$ for $n=0,...,3$ where

$$h(n) = \{\underset{\uparrow}{3}, 2\}$$

$$x(n) = \{\underset{\uparrow}{5}, 4\}$$

and $x(n)$ and $h(n)$ are **zero** for other values of n .

Solution

- From

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k)$$

- If $h(n)$ and $x(n)$ is zero when $0 < n$ and $n > 1$, then we calculate convolution only for $h(0) = 0$ and 1.

$$y(n) = \sum_{k=0}^1 h(k)x(n-k)$$

- Note that, like n , k is just an index for the sum.

Calculate Convolution at $n=0$

$n = 0$ and let $k = -2, -1, 0, 1, 2$

$$\begin{aligned} y(0) &= 3 \sum_{k=-2}^2 \delta(k)x(0-k) + 2 \sum_{k=-2}^2 \delta(k-1)x(0-k) \\ &= 3 \left[\delta(-2)x(2) + \delta(-1)x(1) + \boxed{\delta(0)x(0)} + \delta(1)x(-1) + \delta(2)x(-2) \right] + \\ &\quad 2 \left[\delta(-3)x(2) + \delta(-2)x(1) + \delta(-1)x(0) + \boxed{\delta(0)x(-1)} + \delta(1)x(-2) \right] \\ &= 3x(0) + 2x(-1) \\ &= 3 \cdot 5 + 2 \cdot 0 = 15 \end{aligned}$$

Calculate Convolution at $n=1$

$n = 1$ and let $k = -2, -1, 0, 1, 2$

$$\begin{aligned}y(1) &= 3 \sum_{k=-2}^2 \delta(k)x(1-k) + 2 \sum_{k=-2}^2 \delta(k-1)x(1-k) \\&= 3 \left[\delta(-2)x(3) + \delta(-1)x(2) + \boxed{\delta(0)x(1)} + \delta(1)x(0) + \delta(2)x(-1) \right] + \\&\quad 2 \left[\delta(-3)x(4) + \delta(-2)x(3) + \delta(-1)x(2) + \boxed{\delta(0)x(1)} + \delta(1)x(0) + \delta(2)x(-1) \right] \\&= 3x(1) + 2x(0) \\&= 3 \cdot 4 + 2 \cdot 5 = 22\end{aligned}$$

Calculate Convolution at $n=2$

$n = 2$ and let $k = -2, -1, 0, 1, 2$

$$\begin{aligned} y(2) &= 3 \sum_{k=-2}^2 \delta(k) x(2-k) + 3 \sum_{k=-2}^2 \delta(k-1) x(2-k) \\ &= 3 \left[\delta(-2)x(4) + \delta(-1)x(3) + \boxed{\delta(0)x(2)} + \delta(1)x(1) + \delta(2)x(0) \right] + \\ &\quad 2 \left[\delta(-3)x(4) + \delta(-2)x(3) + \delta(-1)x(2) + \boxed{\delta(0)x(1)} + \delta(1)x(0) \right] \\ &= 3x(2) + 2x(1) \\ &= 3 \cdot 0 + 2 \cdot 4 = 8 \end{aligned}$$

Calculate Convolution at n=3

n = 3 and let $k = -2, -1, 0, 1, 2$

$$\begin{aligned} y(3) &= 3 \sum_{k=-2}^2 \delta(k)x(3-k) + 2 \sum_{k=-2}^2 \delta(k-1)x(3-k) \\ &= 3 \left[\delta(-2)x(5) + \delta(-1)x(4) + \boxed{\delta(0)x(3)} + \delta(1)x(2) + \delta(2)x(1) \right] + \\ &\quad 2 \left[\delta(-3)x(5) + \delta(-2)x(4) + \delta(-1)x(3) + \boxed{\delta(0)x(2)} + \delta(1)x(1) \right] \\ &= 3x(3) + 2x(2) \\ &= 3 \cdot 0 + 2 \cdot 0 = 0 \end{aligned}$$

n	$y(n)$
0	$3x(0) + 2x(-1)$
1	$3x(1) + 2x(0)$
2	$3x(2) + 3x(1)$
3	$3x(3) + 2x(2)$

$$y(n) = 3x(n) + 2x(n-1)$$

Homework 2

- Determine the convolution results of $y(n)=x(n)*h(n)$ for $n=-1,...,3$ where

$$x(n) = [1, \underset{\uparrow}{2}, 3]$$

$$h(n) = [\underset{\uparrow}{1}, 1, 1]$$

- and $x(n)$ and $h(n)$ are zero for other values of n .

Answer

$$\begin{aligned} y(n) &= h(n) * x(n) = x(n) * h(n) \\ &= \sum_{k=-\infty}^{\infty} h(k)x(n-k) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) \\ &= [1, \underset{\uparrow}{3}, 6, 5, 3] \end{aligned}$$

The Properties of Convolution

- Cumulative Property

$$x(n) * h(n) = h(n) * x(n)$$

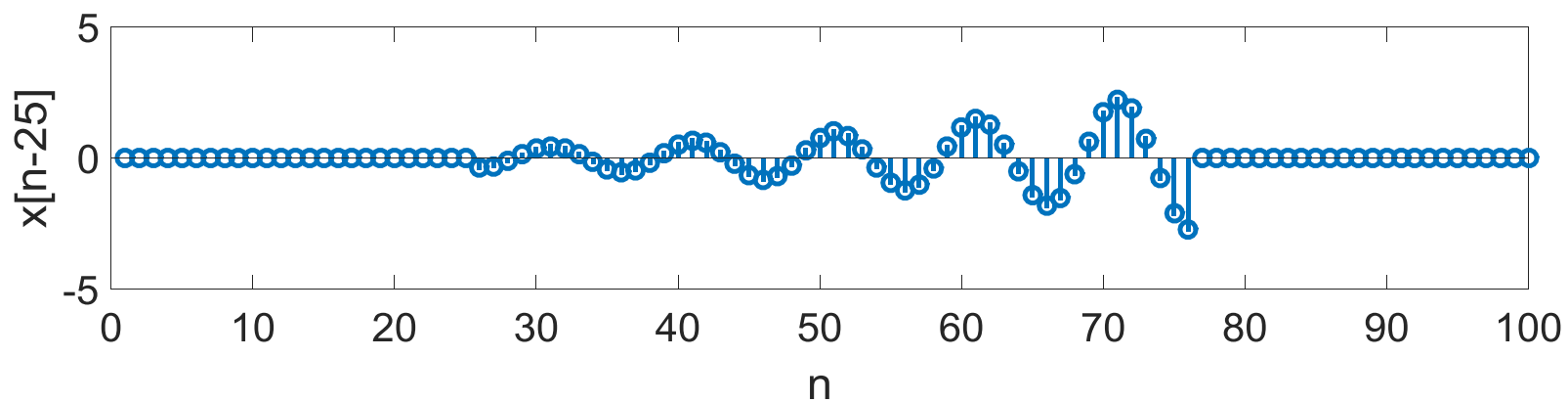
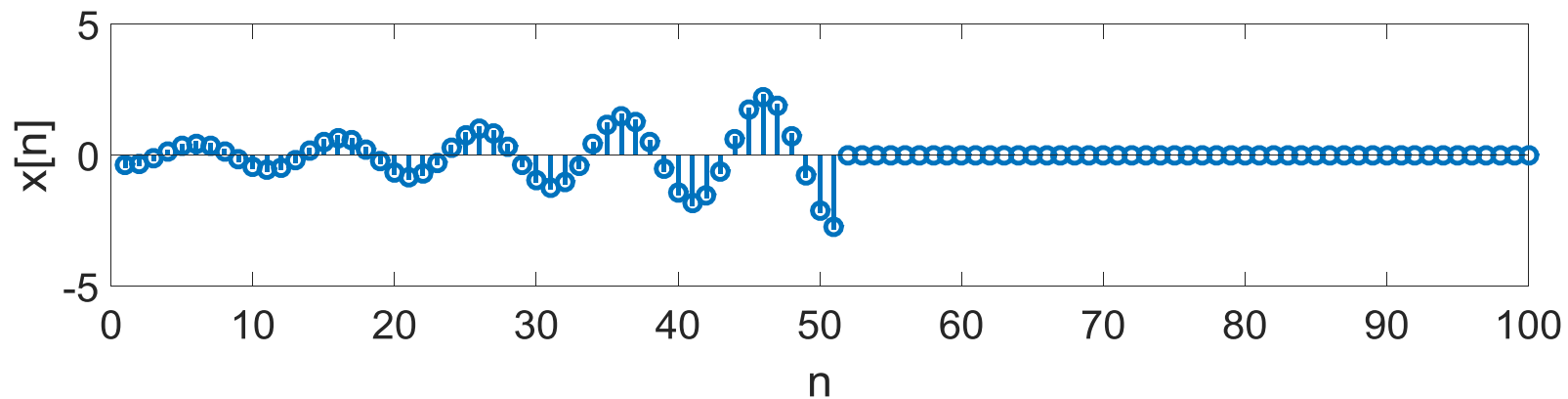
- Associative property

$$\{x(n) * h_1(n)\} * h_2(n) = x(n) * \{h_1(n) * h_2(n)\}$$

- Distributive property

- $$x(n) * \{h_1(n) + h_2(n)\} = x(n) * h_1(n) + x(n) * h_2(n)$$

Time Shifting



Time Reversal

